



RV College of
Engineering®

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Unit-1

AI Foundations for Engineers

CI124TA

■ Introduction

- What is AI ?
- Foundation of Artificial Intelligence
- History of Artificial Intelligence
- The State of the Art

■ Intelligent Agents

- Introduction
- How Agents should Act
- Structure of Intelligent Agents

■ Problem Solving

- Solving the Problems by Searching
- Search Strategies
- Avoiding the Repeated States



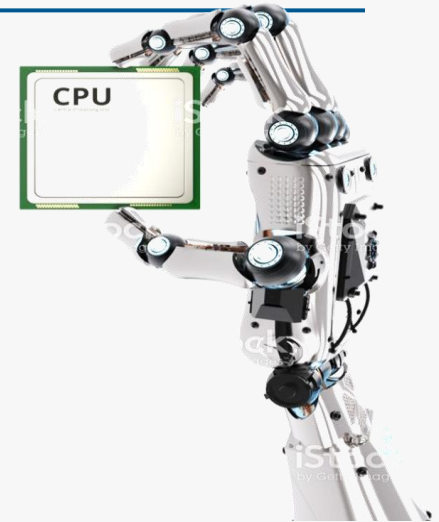
Introduction

What is AI ?

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- Artificial Intelligence (AI) - The ability for a computer to think and learn
 - Perceive - to become aware of (observe)
 - Understand
 - Predict
 - Manipulate - to control, manage - skilfully

- Artificial Intelligence (AI) takes advantage of computers and machines to mimic the problem-solving and decision-making capabilities of human beings
 - Computer: Intelligent enough to process and analyse the data
 - Machine: Contains no intelligence, waiting for instructions from the user or computer



- Definitions of **Artificial Intelligence (AI)** generally differ along **two important dimensions: Thinking** and **Acting**.
- AI Concerned with Thinking (Thought Processes and Reasoning)
- How machines **simulate human thinking and cognitive processes**. The goal is to make computers **think like humans or think rationally**.
- Skills that develop over time in such systems include:
 - **Reasoning** – Drawing logical conclusions from available information.
 - **Problem Solving** – Finding solutions to complex situations.
 - **Judging / Decision Making** – Selecting the best option among alternatives.
 - **Remembering (Memory)** – Storing and retrieving knowledge for future use.
- In this approach, AI attempts to **model human cognitive abilities**, similar to how humans analyze situations and make decisions.

- **2. AI Concerned with Acting (Behavior)**
- The second dimension focuses on **how an AI system behaves or acts in an environment** rather than how it thinks internally.
- An AI system is considered **rational** if it:
 - Does the right thing based on the information it has.
- In other words, the system **chooses actions that maximize the chances of achieving its goals** given its knowledge and situation.

Example: Career Decision

- An employee may choose between:
 - **Staying in the same company**
 - **Changing to a new company**
- The decision might depend on salary, job satisfaction, growth opportunities, or work environment.
- A rational decision would **maximize long-term benefits**.

What is AI ? (Contd.)

1

Thinking Humanly

“The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense.” (Haugeland, 1985)

“[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)

3

Acting Humanly

“The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990)

“The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)

2

Thinking Rationally

“The study of mental faculties through the use of computational models.”

(Charniak and McDermott, 1985)

“The study of the computations that make it possible to perceive, reason, and act.”
(Winston, 1992)

4

Acting Rationally

“Computational Intelligence is the study of the design of intelligent agents.” (Poole *et al.*, 1998)

“AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

What is AI ? (Contd.)

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Feature	Thinking Humanly	Acting Humanly
Focus	Human-like reasoning	Human-like behavior
Goal	Simulating human thought processes	Producing human-like responses/actions
Evaluation	Based on cognitive models, neuroscience	Based on performance (e.g., Turing Test)
Example AI	Cognitive architectures, deep learning	Chatbots, humanoid robots, NPCs

Feature	Thinking Rationally	Acting Rationally
Focus	Logical reasoning	Goal-directed behavior
Approach	Symbolic logic, expert systems	Utility maximization, reinforcement learning
Strengths	Accuracy, formal correctness	Adaptability, real-world effectiveness
Weaknesses	Struggles with uncertainty, impractical for complex tasks	May not always find the "perfect" logical solution

- Definition of AI are categorized into following four categories

1. System that thinks like Humans

2. Systems that think Rationally

3. System that act like Humans

4. Systems that act Rationally

- A **Human-Centered Approach** is an empirical science, involving hypothesis and experimental (observations and experience)
- A **Rationalist Approach** involves a combination of mathematics and engineering.

Rationalist Approach = Mathematics + Engineering

- Generally, all four approaches have been followed, during the development of AI systems

Acting Humanly: The Turing Test Approach

- The Turing Test, proposed by **Alan Turing** (1950), was designed to provide a satisfactory operational definition of intelligence.
- Turing Machine defined to exhibit intelligent behavior as the ability to achieve **human-level performance in all cognitive tasks**, sufficient to fool an interrogator.
- Roughly speaking, the test he proposed is that the computer should be interrogated by a human via a teletype, and passes the test if the interrogator cannot tell if there is a computer or a human at the other end.
- The Turing Machine (Computer) would need to possess the following Human capabilities

Acting Humanly: The Turing Test Approach

- **Natural Language Processing (NLP):** to enable it to communicate successfully in English (or some other human language)
 - **Knowledge Representation:** to store information provided before or during the interrogation;
 - **Automated Reasoning** to use the stored information to answer questions and to draw new conclusions;
 - **Machine Learning (ML)** to adapt to new circumstances and to detect and extrapolate patterns
- Turing's test deliberately avoided direct physical interaction between the interrogator and the computer, because physical simulation of a person is unnecessary for intelligence
 - However, total Turing Test includes a video signal so that the interrogator can test the subject's perceptual abilities, as well as the opportunity for the interrogator to pass physical objects "through the hatch."

Acting Humanly: The Turing Test Approach

- To pass the total Turing Test, the computer will need
 - **Computer Vision** to perceive objects, and
 - **Robotics** to manipulate the objects (move around)
- Acting like a human comes up primarily when AI programs interact with people, as when an expert system explains how it came to its diagnosis, or a natural language processing system has a dialogue with a user.
- These programs must behave according to certain normal conventions of human interaction in order to make themselves understood.

Thinking Humanly: The Cognitive Modelling Approach

- If the program is said to be thinking like human, some ways to determine the human thinking need to be listed
- There are two ways to understand the actual working of the human minds
 - **Through Introspection:** Trying to catch the human thoughts as they go by (Examination of one's own mental and emotional process)
 - **Through Psychological Experiments:** scientific procedure undertaken to make a discovery, testing hypothesis/demonstrating known facts.
- Once we have a sufficiently precise theory of the mind, it becomes possible to express the theory as a **computer program**.
- If the program's input/output and timing behavior matches human behavior, - **cognitive modelling**.

Key Idea

The system should **follow the same reasoning steps that a human would follow.**

Example:

A human solving a puzzle might:

1. Observe the problem
2. Recall previous experience
3. Apply logical reasoning
4. Test possible solutions

A **thinking humanly AI** attempts to replicate these steps.

Thinking Humanly: The Cognitive Modelling Approach

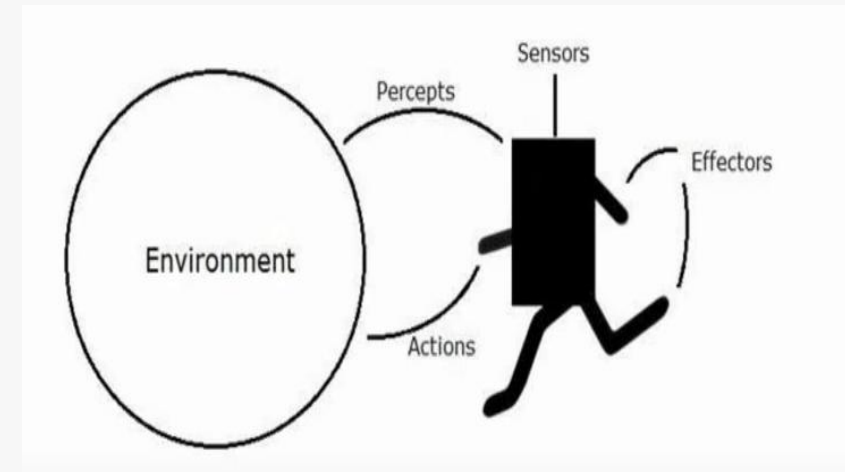
- The interdisciplinary field of **Cognitive Science** brings together **Computer Models from AI** and experimental techniques from **Psychology** to try to construct precise and testable theories of the workings of the human mind.
- Real Cognitive Science, is based on experimental investigation of actual humans or animals
- AI and Cognitive Science continue to fertilize each other, especially in the areas of vision, natural language, and learning

⇒ Thinking Rationally: Correct Reasoning

- If a machine wants to think rightly it needs: 100% knowledge and so many computations.
- If something is true, then that must be true.
- If someone is thinking correctly / rightly in a given circumstances in a given amount of information - **The law of thought Approach**
- Example of logical reasoning:
 - All humans are mortal
 - Socrates is a human
 - Therefore, Socrates is mortal
 - This type of reasoning is called **deductive logic**.

Acting rationally: The rational agent approach

- An **Agent** is just something that Perceives(realize/understand) and acts
- Acting Rationally: Doing right things
- In this approach, AI is viewed as the study and construction of rational agents
- AI agent is anything that perceives its environment through sensor and acts upon that environment through effectors.



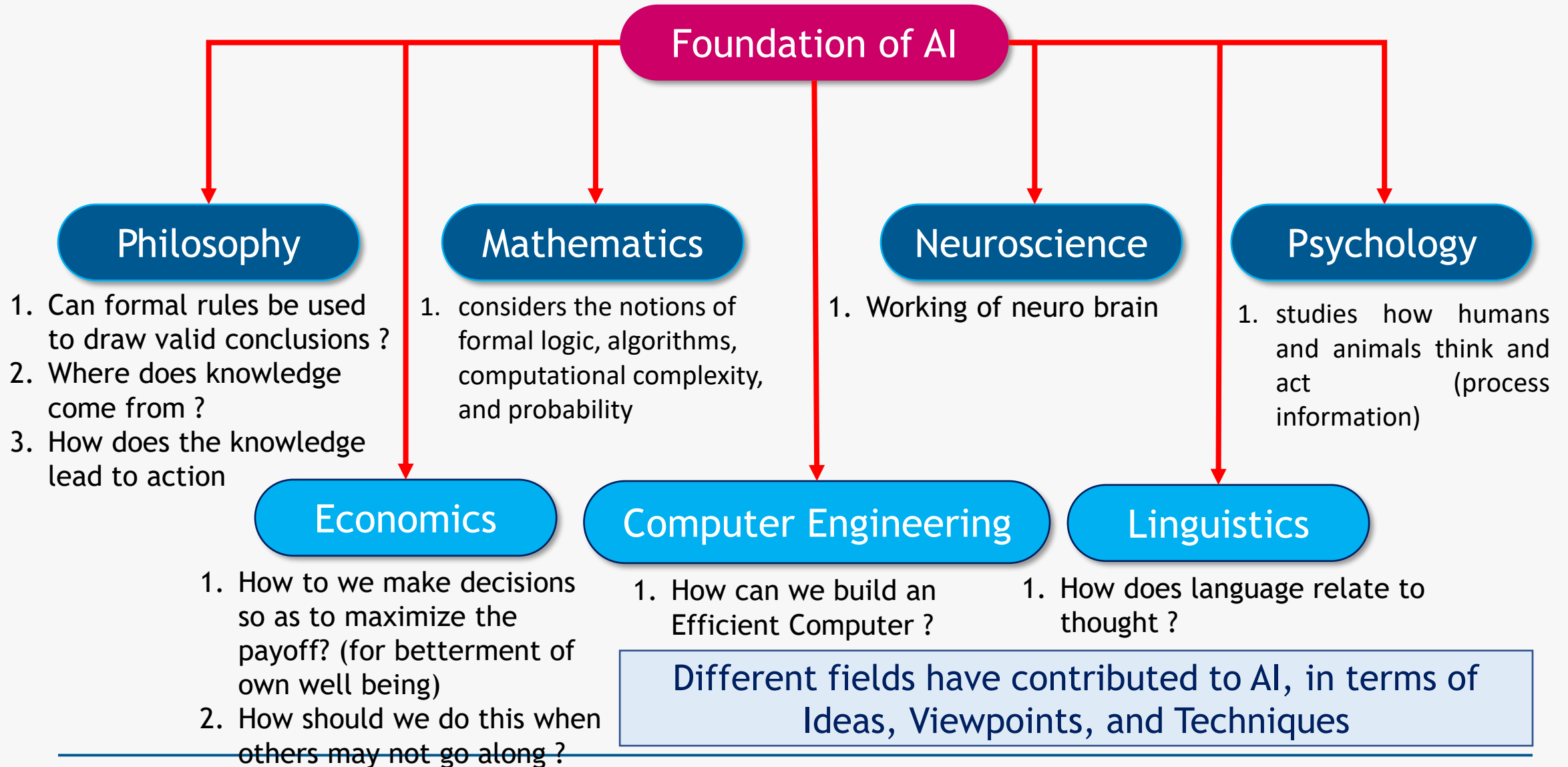
Acting rationally: The Rational Agent Approach

- On the other hand, correct inference is not all of rationality, because there are often situations where there is no provably correct thing to do, yet something must still be done
- There are also ways of acting rationally that cannot be reasonably said to involve inference. For example, pulling one's hand off of a hot stove is a reflex action that is more successful than a slower action taken after careful deliberation
- To allow Rational Actions, all of the “Cognitive Skills” required for the Turing Test are present.
- Thus, we need the ability to represent knowledge and reason with it because this enables us to reach good decisions in a wide variety of situations.

Acting rationally: The Rational Agent Approach

- One important point to keep in mind: Achieving perfect rationality—always doing the right thing—is not possible in complicated environments. The computational demands are just too high.
- Limited Rationality- Acting appropriately when there is not enough time to do all the computations one might like.

- This section provide a brief history of AI
- Although AI itself is a young field, it has inherited many ideas, viewpoints, and techniques from other disciplines.
 - From over 2000 years of tradition in philosophy, theories of reasoning and learning have emerged, along with the viewpoint that the mind is constituted by the operation of a physical system.
 - From over 400 years of mathematics, which comprises of formal theories of logic, probability, decision making, and computation.
 - From psychology, the tools to investigate the human mind, and a scientific language to express the resulting theories. From linguistics, the theories of the structure and meaning of language.
 - Finally, from computer science, the tools with which to make AI a reality.



1. Saves Time and Effort

AI can perform repetitive tasks much faster than humans. It helps automate activities like **data entry, scheduling, and answering customer queries.**

2. Improves Accuracy

AI reduces human errors in tasks such as medical diagnosis, financial analysis, and data processing.

1. *Example:* Hospitals use AI to analyze **X-rays or MRI scans** to detect diseases earlier than humans.

Personalization: AI customizes experiences based on user behavior.

Example: Platforms like **Netflix** and **Spotify** recommend movies or songs based on your preferences.



Enhances Learning and Research: AI helps students understand complex concepts quickly, generate summaries, and assist in coding or research.

-
- 1. Overdependence on AI** People may rely too much on AI and stop thinking critically.
 - 2. Job Displacement** Automation can replace some human jobs, especially repetitive ones.
 - 3. Privacy and Data Security Issues:** AI systems collect huge amounts of personal data, which can lead to privacy concerns if misused.
 - 4. Bias and Incorrect Information** AI may produce biased or wrong outputs if trained on poor data.

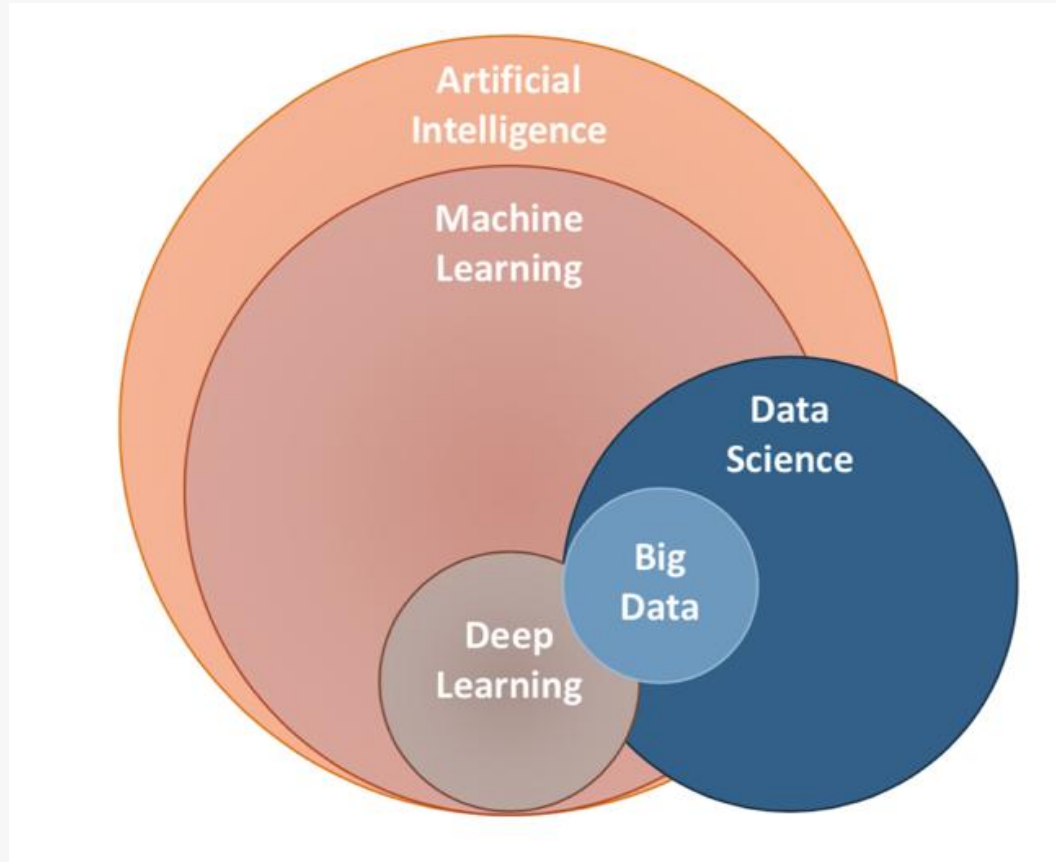
Human Learning

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- Process of gaining information through observation to perform some task and improve the performance in performing the task.
- Why?
- homework, rocket launch angle, walking on street, new course
- Types
 - Learning under expert guidance
 - Learning guided by knowledge gained from experts
 - Learning by self

What is Machine Learning *Go, change the world[®]*

- Machine Learning is Interdisciplinary course applicable to wide variety of domains .



What is Machine Learning *Go, change the world[®]*

A computer program is said to **learn** from **experience E** with respect to some **class of tasks T** and performance **measure P** , if its *performance at tasks in T , as measured by P , improves with experience E .*

Tom Mitchell

A checkers learning problem:

Task T : playing checkers

Performance measure P : percent of games won against opponents

Training experience E : playing practice games against itself

Fruit Prediction Problem

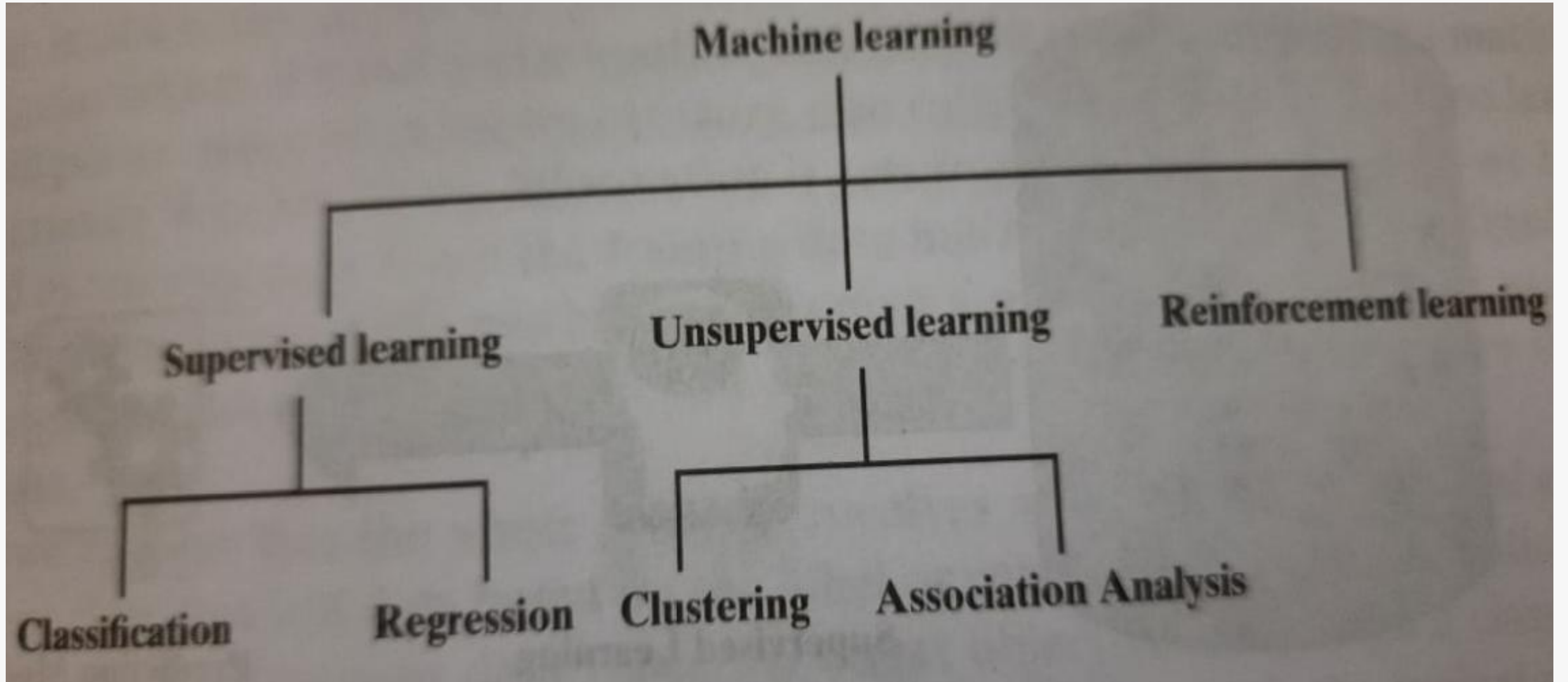
Task – forecasting different fruits for recognition

Performance Measure – able to predict maximum variety of fruits

Experience – training machine with **the largest datasets of fruits images**

Types of Machine Learning

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UNIT - III

- Supervised Learning
- Decision Tree Classifier
- Model Overfitting
- Model Selection

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Supervised Learning

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- Basic Concepts
- General Framework for Classification

Supervised Learning – Basic Concepts

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Classification: Definition

- **Given a collection of records (training set)**
 - **Each record is characterized by a tuple (x,y) , where x is the attribute set and y is the class label**
 - ◆ **x : attribute, predictor, independent variable, input**
 - ◆ **y : class, response, dependent variable, output**
- **Task:**
 - **Learn a model that maps each attribute set x into one of the predefined class labels y**

Supervised Learning – Basic Concepts

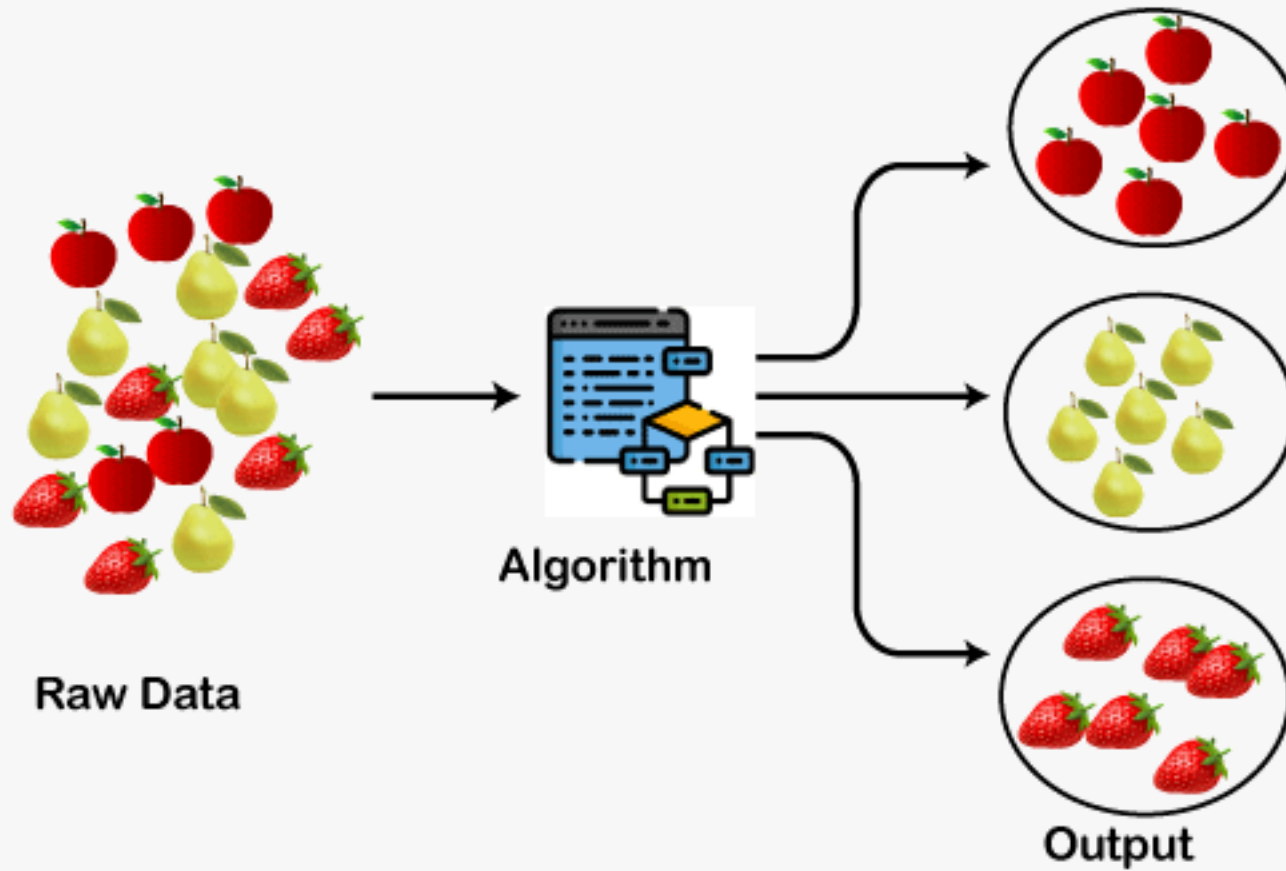
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Examples of Classification

Task	Attribute set, x	Class label, y
Categorizing email messages	Features extracted from email message header and content	spam or non-spam
Identifying tumor cells	Features extracted from x-rays or MRI scans	malignant or benign cells
Cataloging galaxies	Features extracted from telescope images	Elliptical, spiral, or irregular-shaped galaxies

Unsupervised Learning - Clustering

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Unsupervised Learning – Association Analysis

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ID	Items
1	{Bread, Milk}
2	{Bread, Diapers , Beer , Eggs}
3	{Milk, Diapers , Beer , Cola}
4	{Bread, Milk, Diapers , Beer }
5	{Bread, Milk, Diapers, Cola}
...	...

market
basket
transactions

{Diapers, Beer}

Example of a frequent itemset

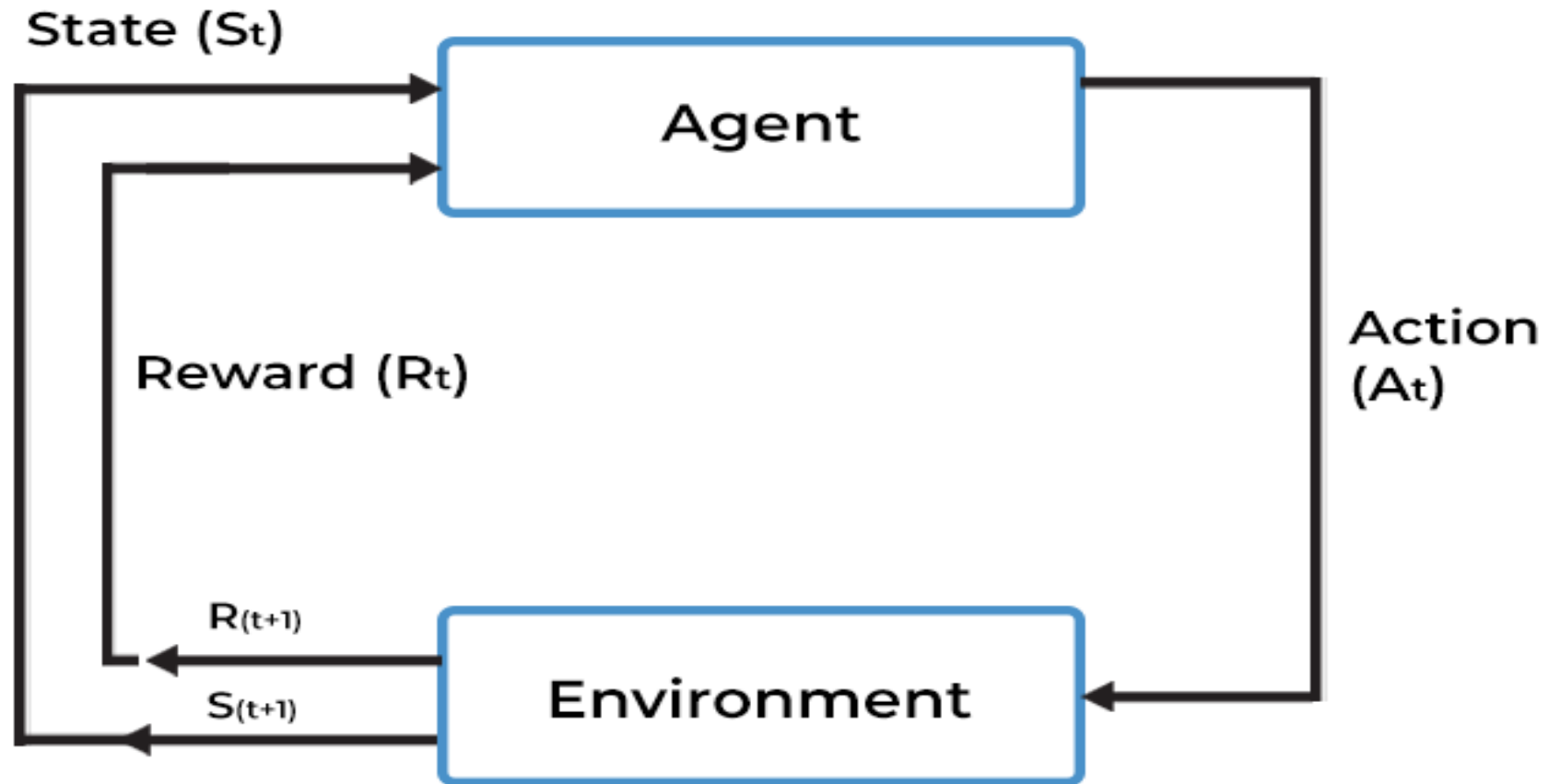
{Diapers} → {Beer}

Example of an association rule

Reinforcement Learning

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REINFORCEMENT LEARNING MODEL



Comparison of various learnings

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Supervised machine learning	Unsupervised machine learning	Reinforcement
Used for classifying (classes and labels are given)	To find pattern (No idea about class/labels)	Rewards – if guessed correctly (No idea about class/labels)
Labelled training data	Unknown and unlabeled data	Model learns and updates itself through rewards/punishment
Classification and regression	Clustering and association analysis	No such types
Simple to understand	More difficult to understand and implement	Most complex
Navie bayes, KNN, LR, SVM etc	Kmeans, PCA, DBSCAN, apriori	Q-Learning, Sarsa
Hand writing recognition, stock market prediction, disease prediction, fraud detection	Market based analysis, recommender system, customer segmentation	Self driving cars, intelligent robots

Problems which maynot be solved using ML

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1. **Critical applications where human are effective – Air traffic control**
2. **Simple applications – price calculator, dispute tracking application**
3. **To an already optimized business process**
4. **Insufficient training data**

Languages and Tools

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1. Python
2. R
3. Matlab
4. SAS
5. SPSS
6. Julia
7. Google colab
8. Jupiter notebook

Descriptive and Predictive Modeling

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- **Descriptive**: determine the similarities in the data and to find existing patterns.
- **Examples of descriptive**: Clustering, association rule mining, and anomaly detection.
 - **Clustering** involves grouping similar objects together
 - **Association rule** mining involves identifying relationships between different items in a dataset.
 - **Anomaly detection** involves identifying unusual patterns or outliers in the data.

Descriptive and Predictive Modeling

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- **Predictive**: It uses the supervised learning functions which are used to predict the target value.
- **Examples**: Regression, Decision tress, and Neural Networks
 - **Regression** involves predicting a continuous outcome variable based on one or more predictor variables
 - **DT** involves building a tree – like model based on set of rules for prediction.
 - **Neural Networks** involves building the model based on human brain to make predictions.

General Approach for Building Classification Model

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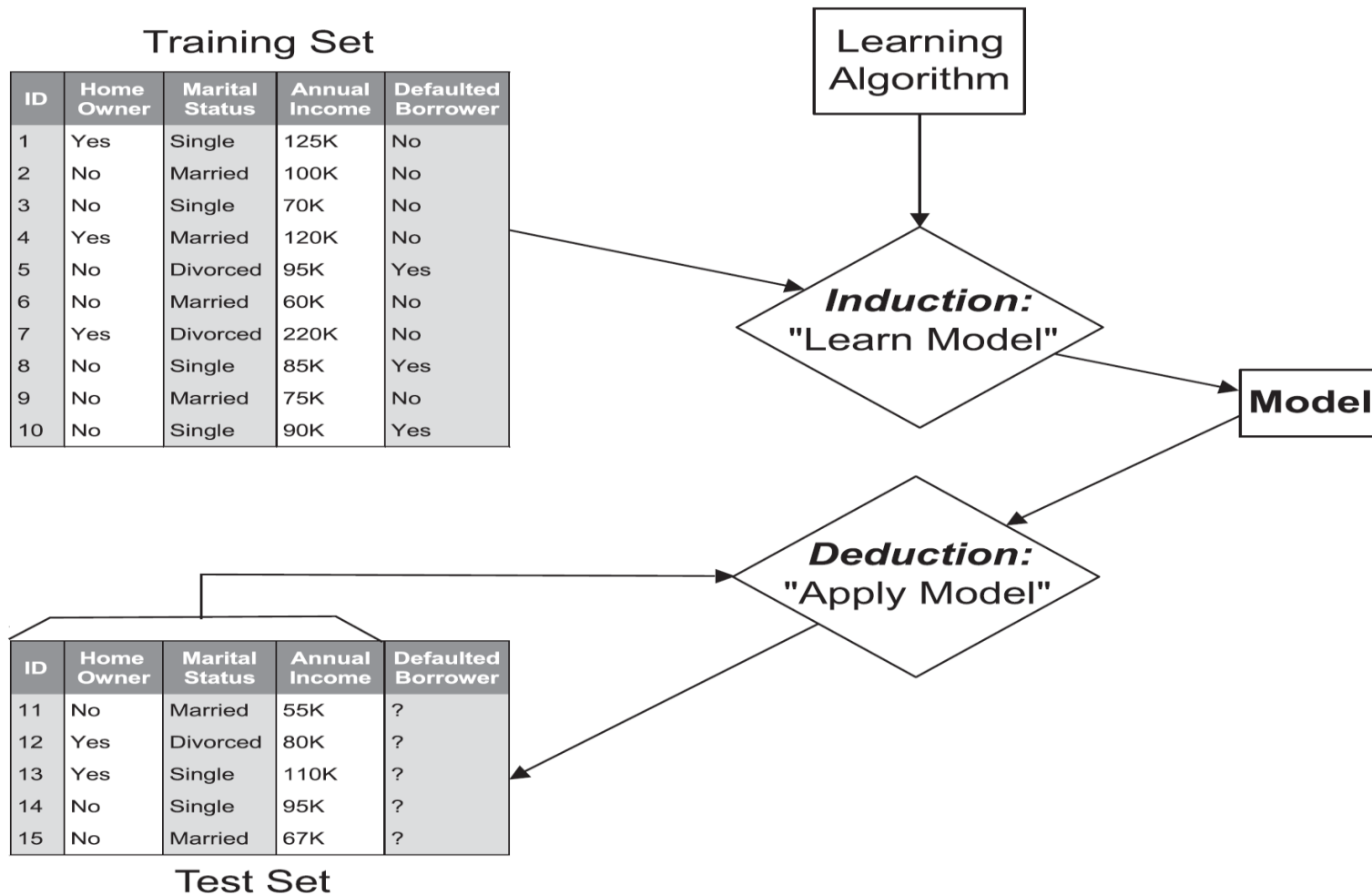
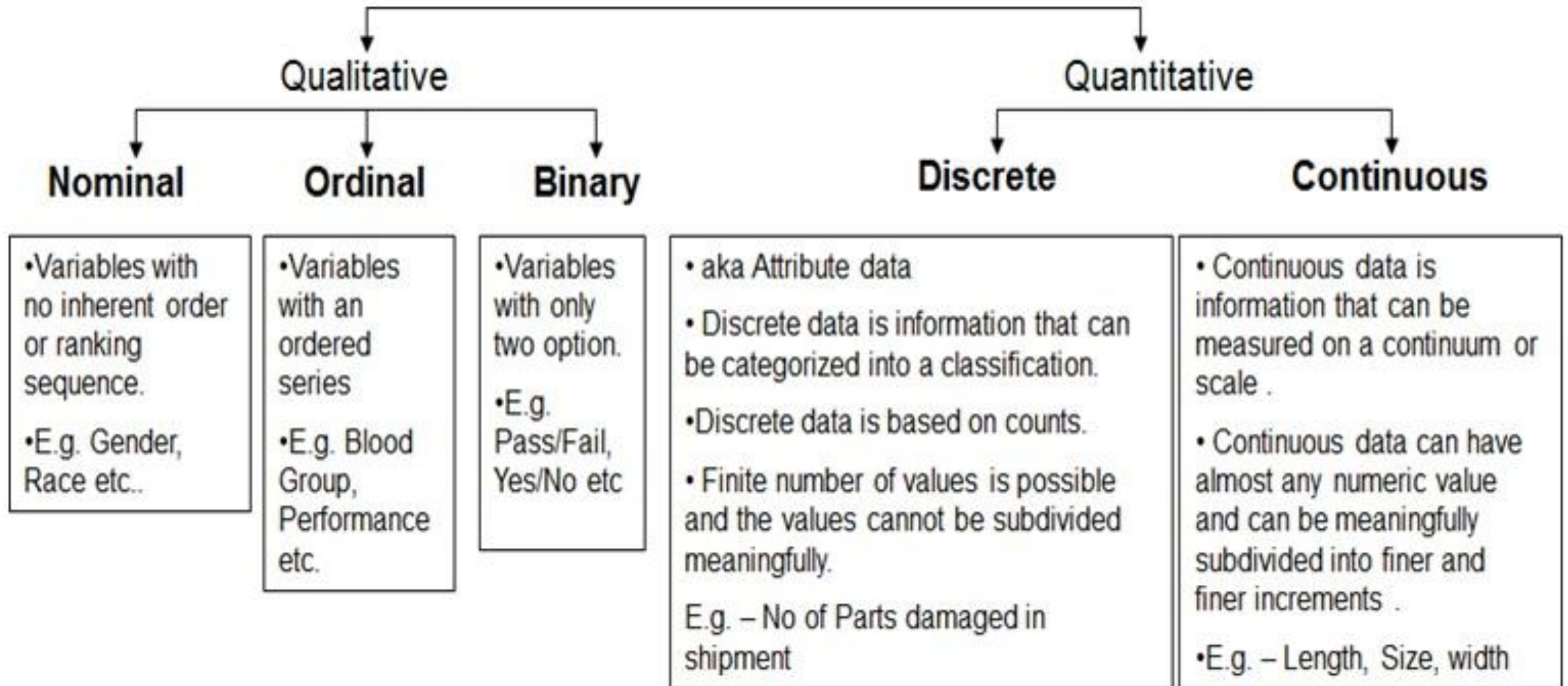


Figure 3.3. General framework for building a classification model.

- Classification techniques are most suited for predicting or describing data sets with **binary or nominal categories**.
- They are **less effective** for **ordinal categories** (e.g., to classify a person as a member of high-, medium-, or low- income group)

Data Attributes

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Classification Techniques

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- **Base Classifiers**

- Decision Tree based Methods
- Rule-based Methods
- Nearest-neighbor
- Naïve Bayes and Bayesian Belief Networks
- Support Vector Machines
- Neural Networks, Deep Neural Nets

- **Ensemble Classifiers**

- Boosting, Bagging, Random Forests

Decision Tree Classifier

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- A Basic Algorithm to Build a Decision Tree
- Methods for Expressing Attribute Test Conditions
- Measures for Selecting an Attribute Test Condition
- Algorithm for Decision Tree Induction
- Characteristics of Decision Tree Classifiers

Decision Tree Classifier

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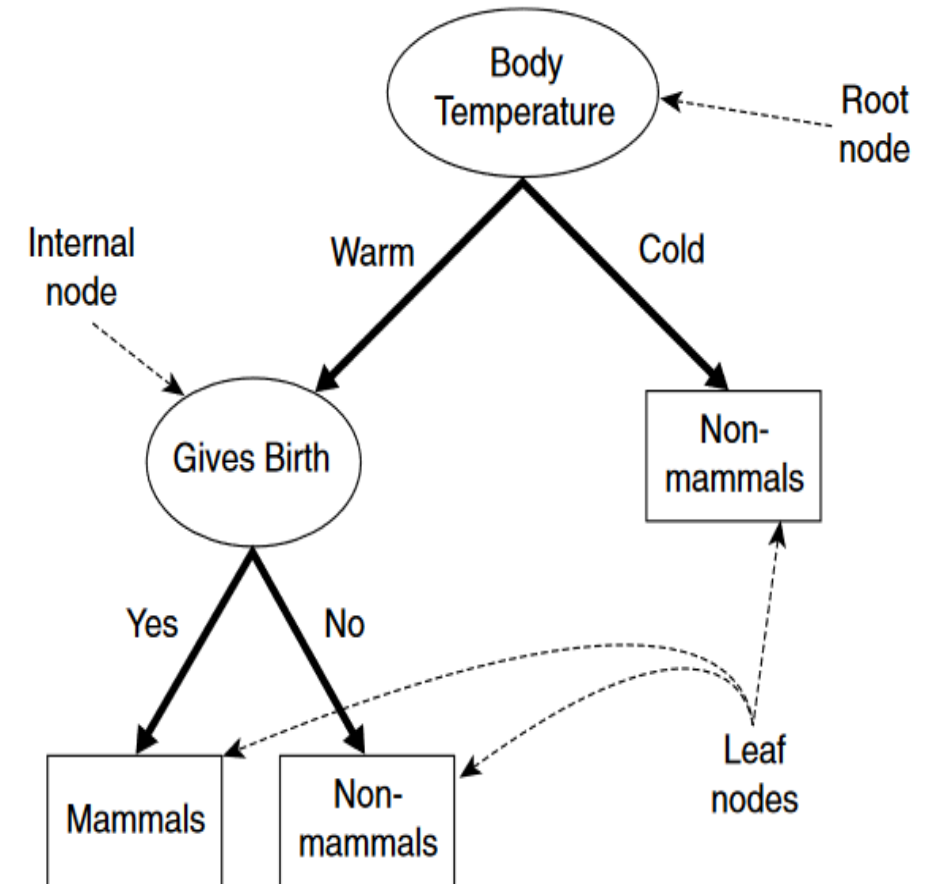
Name	Body Temperature	Skin Cover	Gives Birth	Aquatic Creature	Aerial Creature	Has Legs	Hibernates	Class Label
human	warm-blooded	hair	yes	no	no	yes	no	mammal
python	cold-blooded	scales	no	no	no	no	yes	reptile
salmon	cold-blooded	scales	no	yes	no	no	no	fish
whale	warm-blooded	hair	yes	yes	no	no	no	mammal
frog	cold-blooded	none	no	semi	no	yes	yes	amphibian
komodo dragon	cold-blooded	scales	no	no	no	yes	no	reptile
bat	warm-blooded	hair	yes	no	yes	yes	yes	mammal
pigeon	warm-blooded	feathers	no	no	yes	yes	no	bird
cat	warm-blooded	fur	yes	no	no	yes	no	mammal
leopard	cold-blooded	scales	yes	yes	no	no	no	fish
shark								
turtle	cold-blooded	scales	no	semi	no	yes	no	reptile
penguin	warm-blooded	feathers	no	semi	no	yes	no	bird
porcupine	warm-blooded	quills	yes	no	no	yes	yes	mammal
eel	cold-blooded	scales	no	yes	no	no	no	fish
salamander	cold-blooded	none	no	semi	no	yes	yes	amphibian

The series of questions and their possible answers can be organized in the form of a decision tree

Decision Tree Classifier - Nodes

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- **Root Node:** The initial node at the beginning of a decision tree, where the entire population or dataset starts dividing based on various features or conditions.
- **Decision Nodes:** Nodes resulting from the splitting of root nodes are known as decision nodes. These nodes represent intermediate decisions or conditions within the tree.
- **Leaf Nodes:** Nodes where further splitting is not possible, often indicating the final classification or outcome. Leaf nodes are also referred to as terminal nodes.
- **Sub-Tree:** Similar to a subsection of a graph being called a sub-graph, a sub-section of a decision tree is referred to as a sub-tree. It represents a specific portion of the decision tree.
- **Pruning:** The process of removing or cutting down specific nodes in a decision tree to prevent overfitting and simplify the model.
- **Branch / Sub-Tree:** A subsection of the entire decision tree is referred to as a branch or sub-tree. It represents a specific path of decisions and outcomes within the tree.
- **Parent and Child Node:** In a decision tree, a node that is divided into sub-nodes is known as a parent node, and the sub-nodes emerging from it are referred to as child nodes. The parent node represents a decision or condition, while the child nodes represent the potential outcomes or further decisions based on that condition.

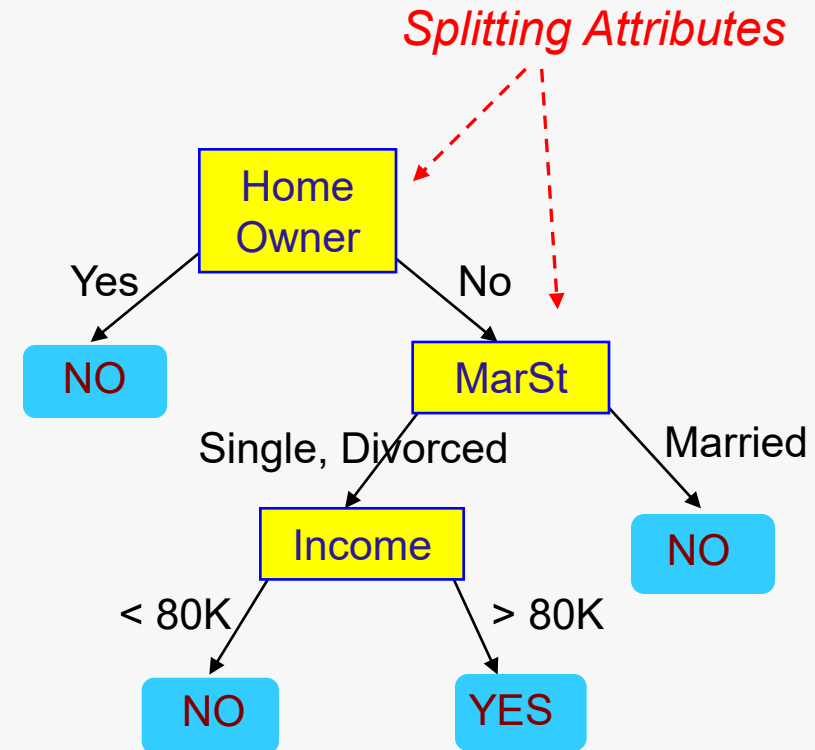


A decision tree for the mammal classification problem

Example of a Decision Tree

ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

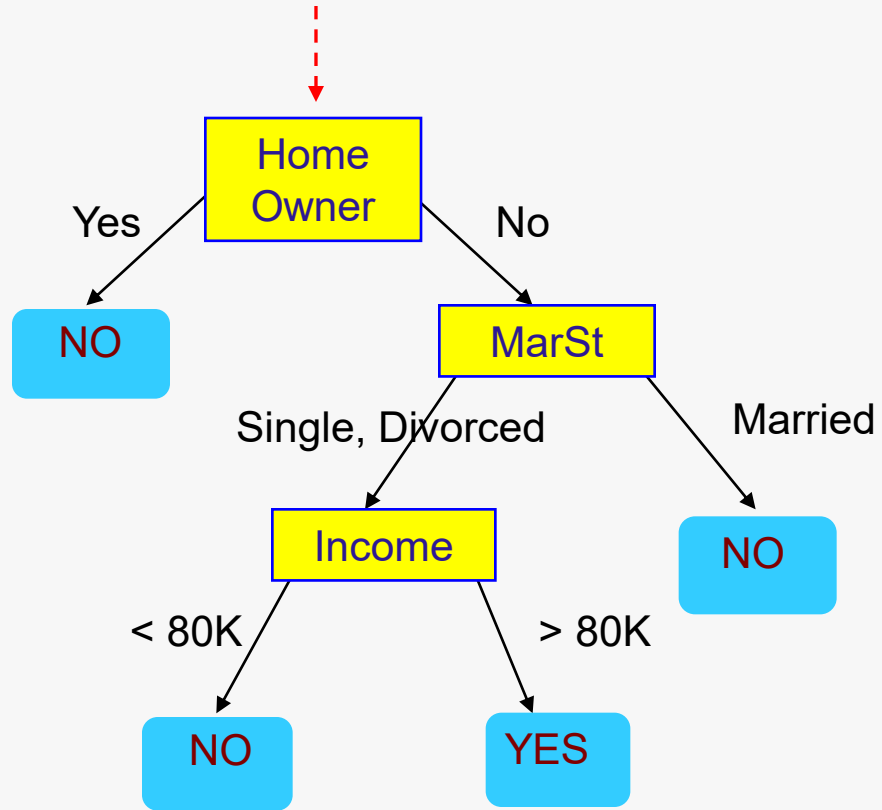
Training Data



Model: Decision Tree

Apply Model to Test Data

Start from the root of tree.



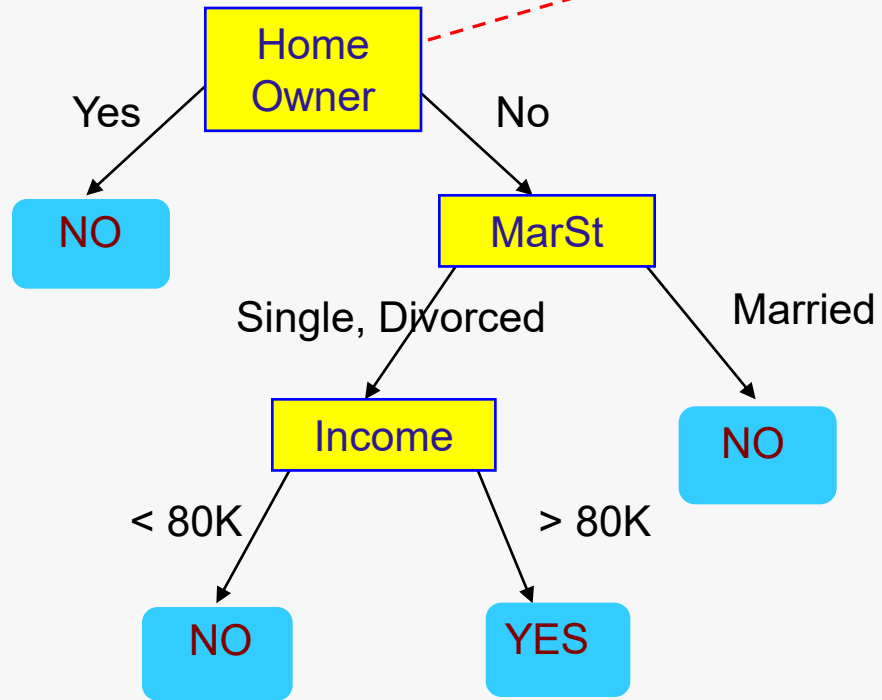
Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

Apply Model to Test Data

Test Data

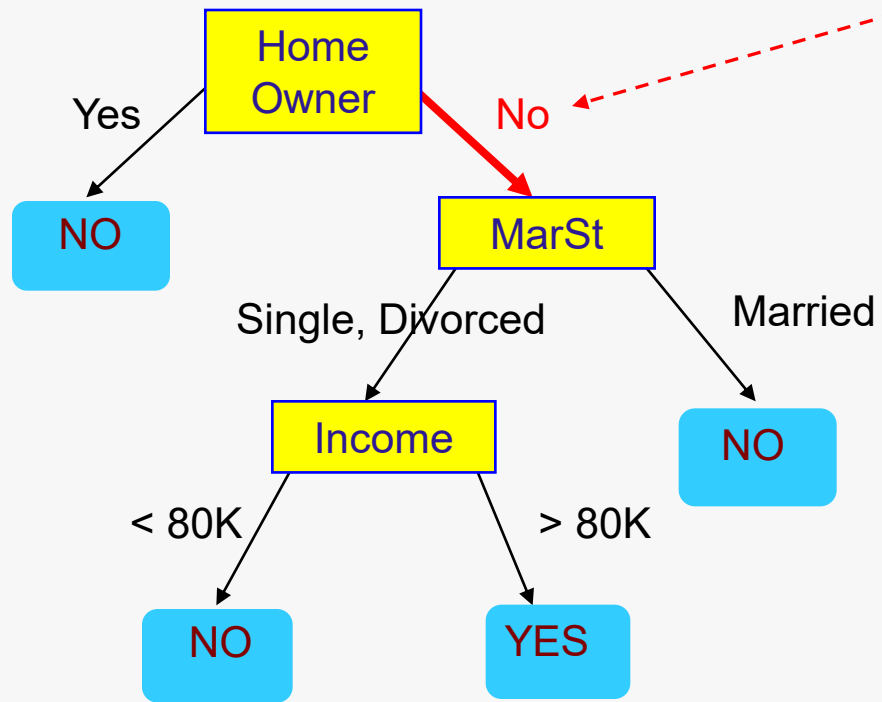
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Apply Model to Test Data

Test Data

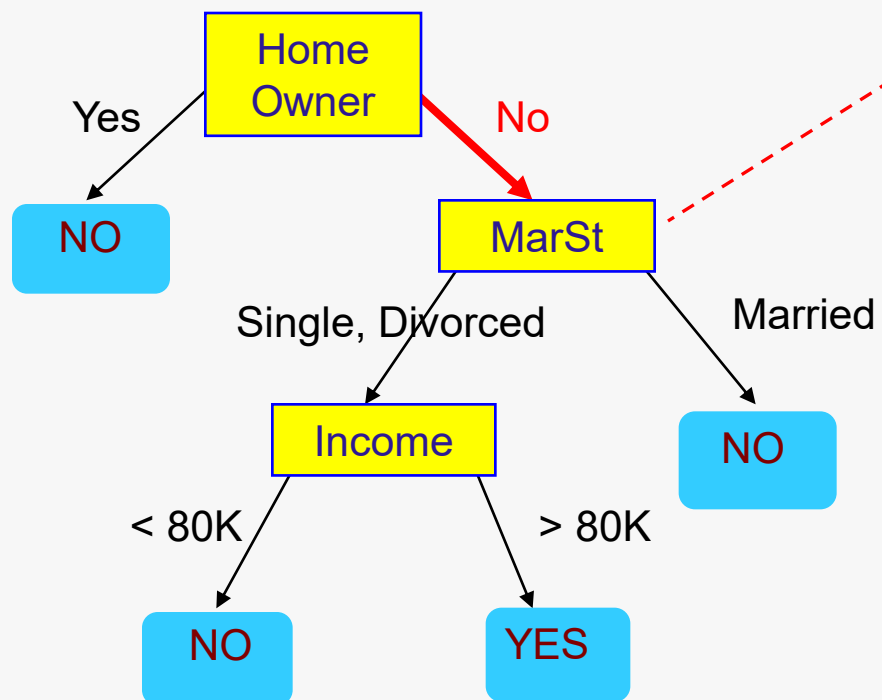
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Apply Model to Test Data

Test Data

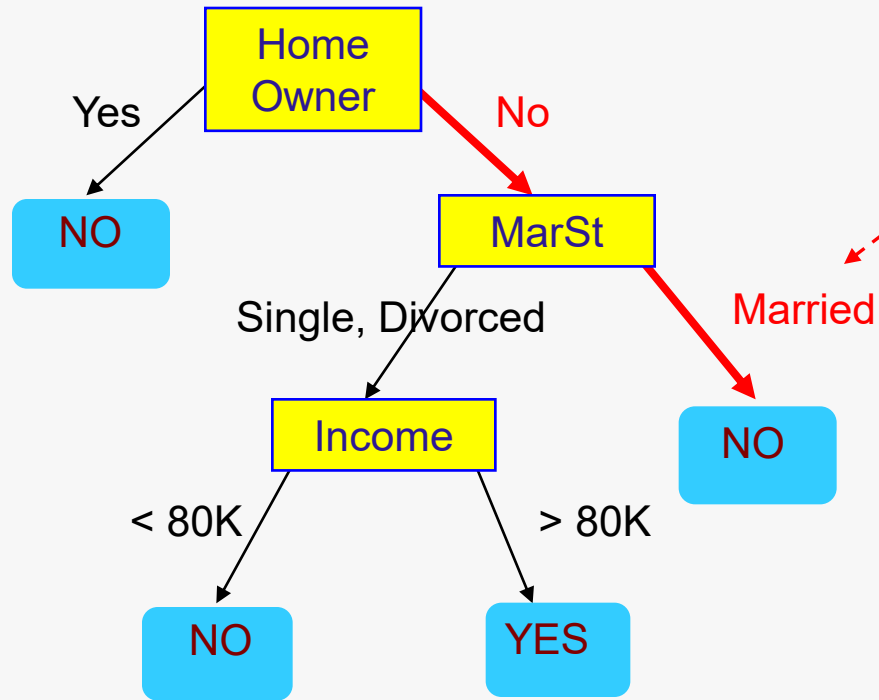
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Apply Model to Test Data

Test Data

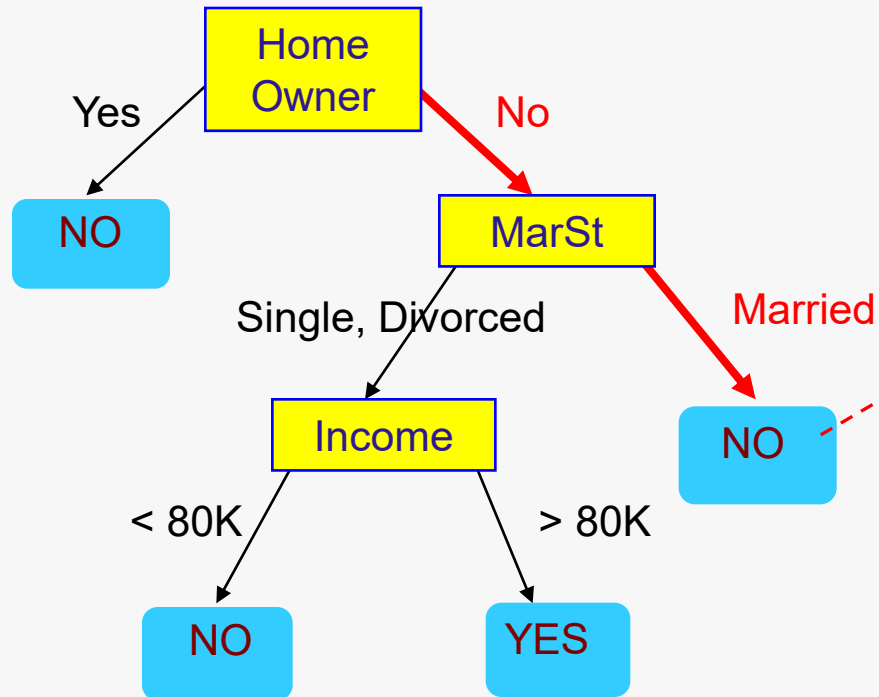
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Apply Model to Test Data

Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Assign Defaulted to
"No"

Another Example of Decision Tree

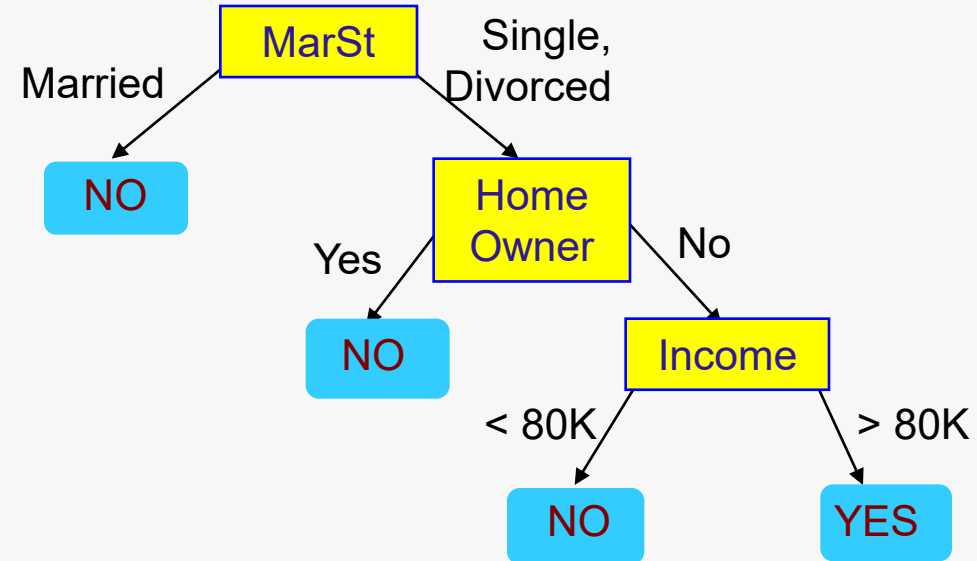
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ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

categorical

categorical

continuous
class



There could be more than one tree that fits the same data!

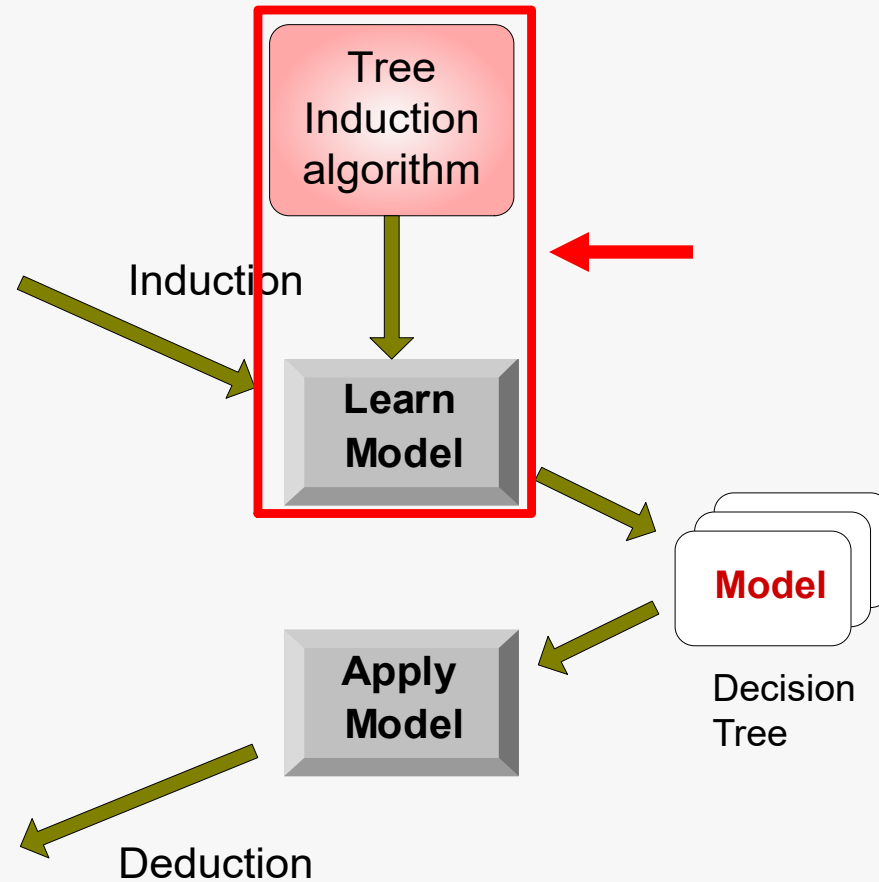
Decision Tree Classification Task *Go, change the world*

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



- **Many Algorithms:**
 - Hunt's Algorithm (one of the earliest)
 - CART
 - ID3, C4.5
 - SLIQ, SPRINT

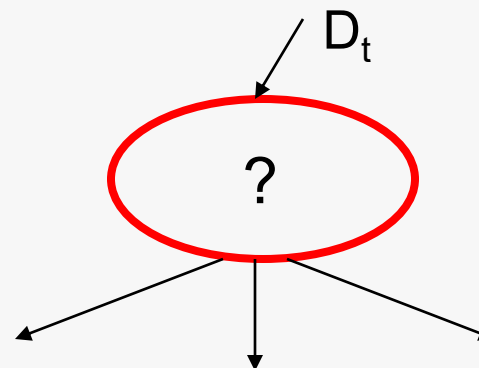
General Structure of Hunt's Algorithm

❓ Let D_t be the set of training records that reach a node t

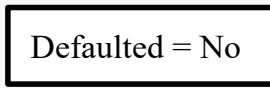
❓ General Procedure:

- If D_t contains records that belong the same class y_t , then t is a leaf node labeled as y_t
- If D_t contains records that belong to more than one class, use an attribute test to split the data into smaller subsets. Recursively apply the procedure to each subset.

ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

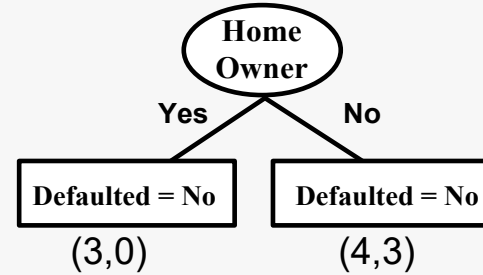


Hunt's Algorithm

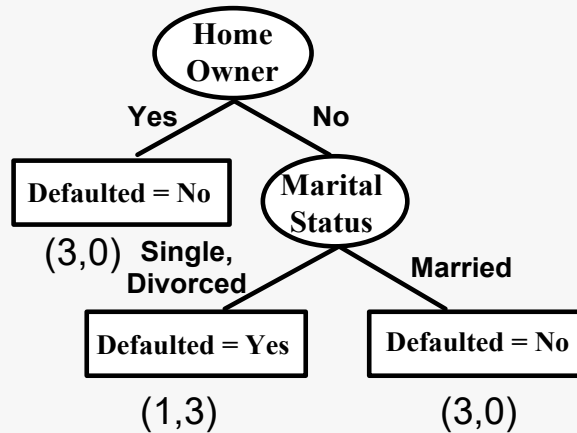


(7,3)

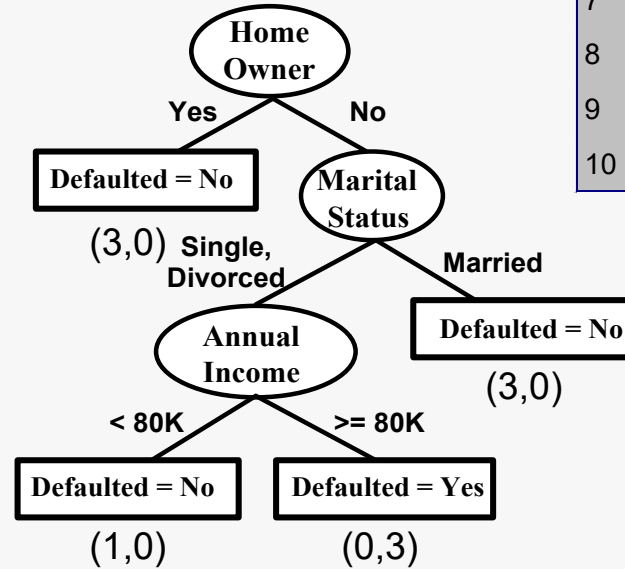
(a)



(b)



(c)



(d)

ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Design Issues of Decision Tree Induction

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❓ How should training records be split?

- Greedy strategy: **split the records** based on some **attribute test** (always choose immediate best option)
- Need to evaluate the “**goodness**” of each attribute test and select the best one.

❓ How should the splitting procedure stop?

- Stop splitting if all the records belong to the **same class or have identical attribute values**
- **Early termination**

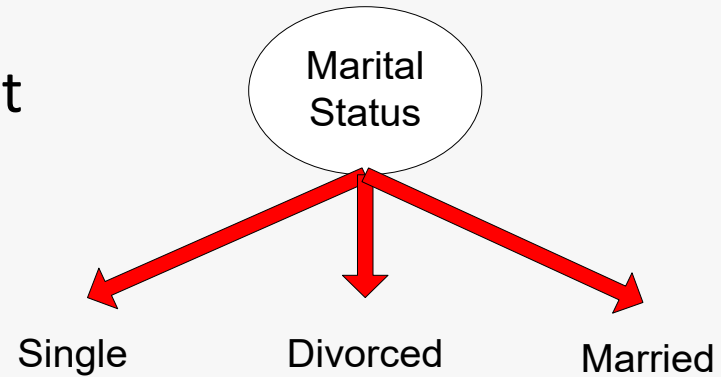
☐ Depends on attribute types

- Binary
- Nominal
- Ordinal
- Continuous

Test Condition for Nominal Attributes

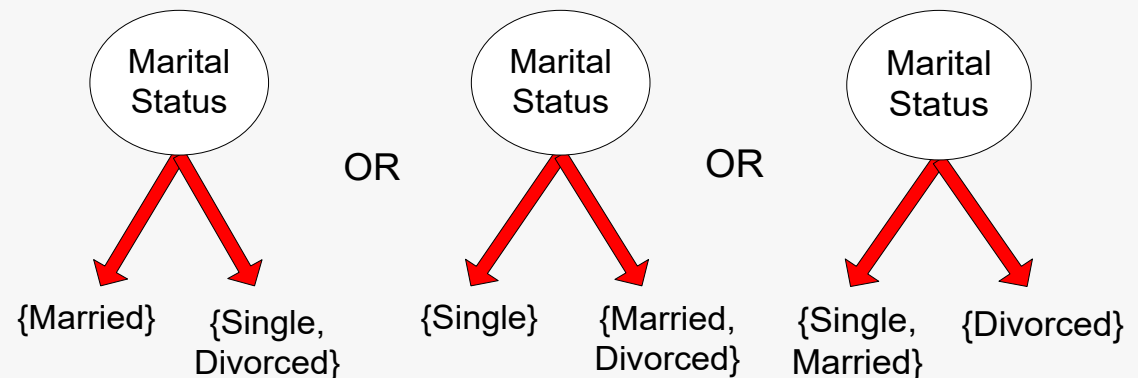
□ Multi-way split:

- Use as many partitions as distinct values.



□ Binary split:

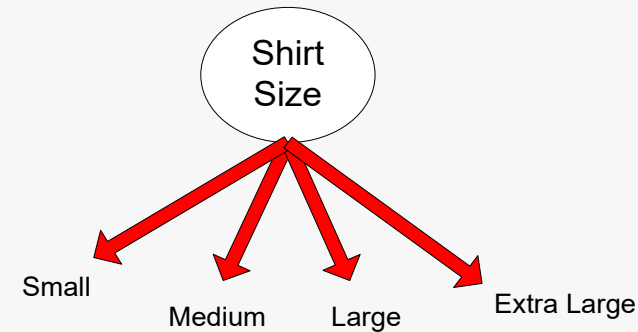
- Divides values into two subsets



Test Condition for Ordinal Attributes

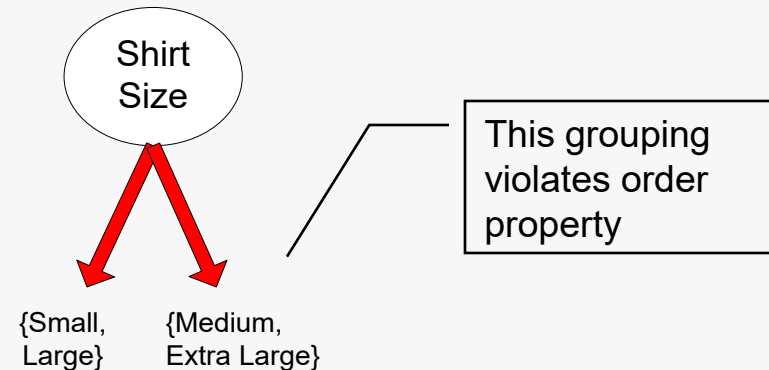
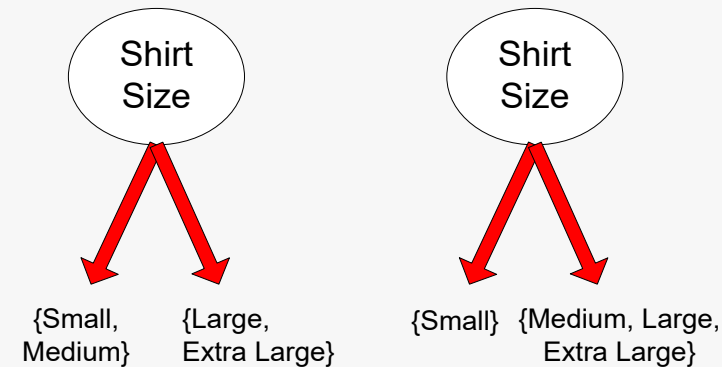
I Multi-way split:

- Use as many partitions as distinct values

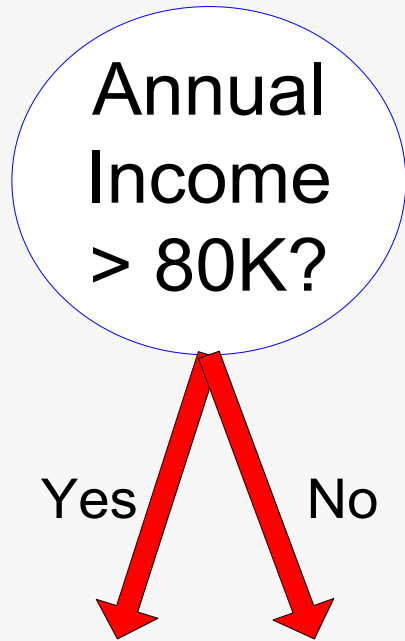


I Binary split:

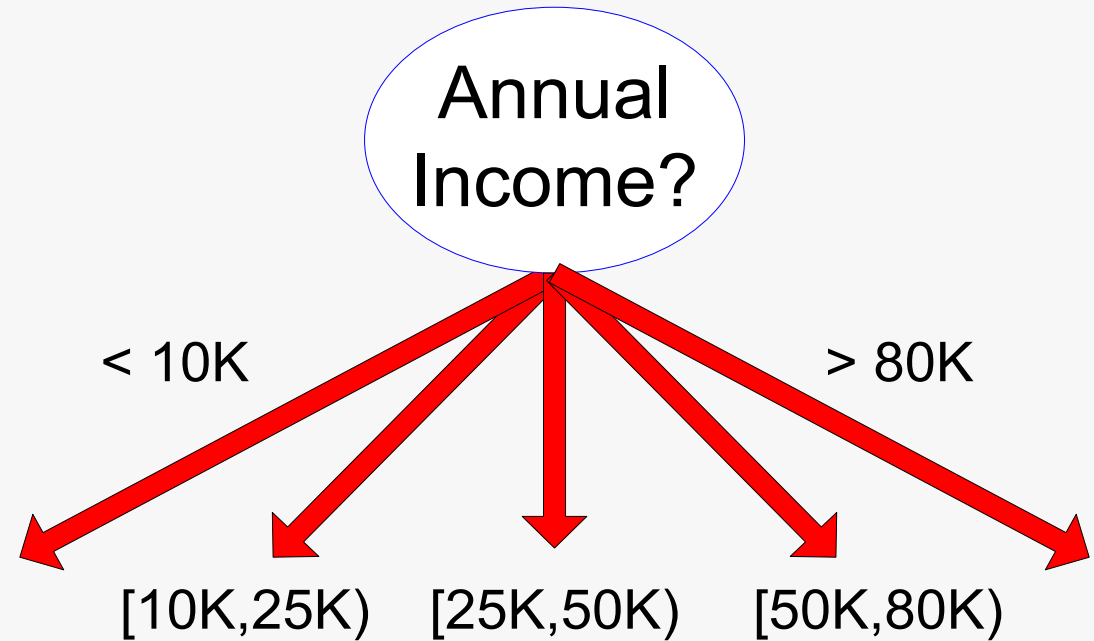
- Divides values into two subsets
- Preserve order property among attribute values



Test Condition for Ordinal Attributes



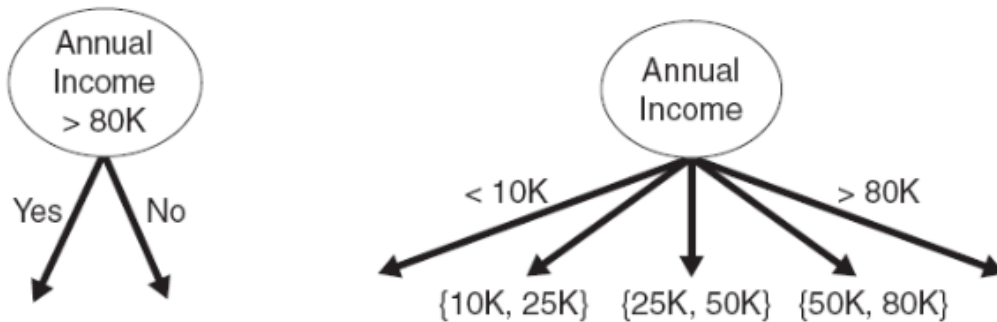
(i) Binary split



(ii) Multi-way split

Test Condition for Continuous Attributes

Splitting Based on Continuous Attributes



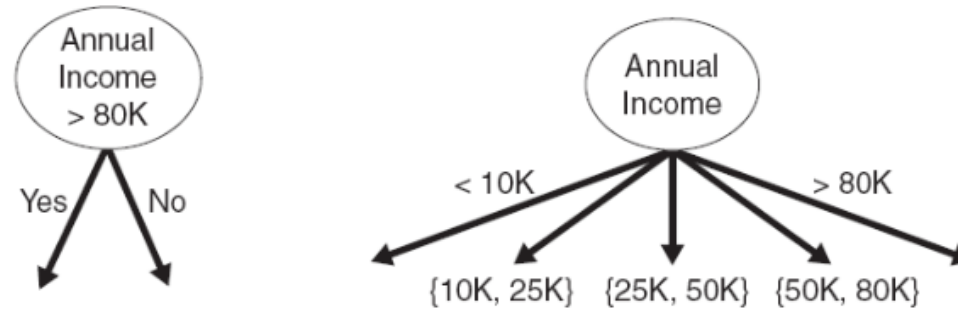
Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Continuous attributes can also have a binary or multiway split.

- Binary: decision tree algorithm must consider all possible split positions v , and it selects the best one
 - Comparison test: $(A \leq v)$ or $(A > v)$, where $v=80K$
- Computationally intensive

Split Condition for Continuous Attributes

Splitting Based on Continuous Attributes



- Continuous attributes can also have a binary or multiway split.
 - ▣ *Multiway*: outcomes of the form $v_i \leq A < v_{i+1}$, for $i = 1, \dots, k$
 - Consider all possible ranges of continuous variable?
 - Use same binning strategies as discussed for preprocessing a continuous attribute into a discrete one
 - ▣ Note: adjacent intervals/“bins” can always be aggregated into wider ones

Split Condition for Continuous Attributes

Splitting for Ordinal Attributes

Definition: Ordinal attributes have a natural order (e.g., **Education Level: High School < Bachelor's < Master's < PhD**).

Binary Splitting (Threshold-Based)

- Treat ordinal data **like continuous data** by converting it into a **binary decision**.
- Example:
 - Suppose the feature "**Education Level**" has 4 categories: **High School, Bachelor's, Master's, PhD**.
 - Possible binary splits:
 - "**Education $\leq$ Bachelor's**" vs. "**Education > Bachelor's**"
 - "**Education $\leq$ Master's**" vs. "**Education > Master's**"

Binning or Discretization

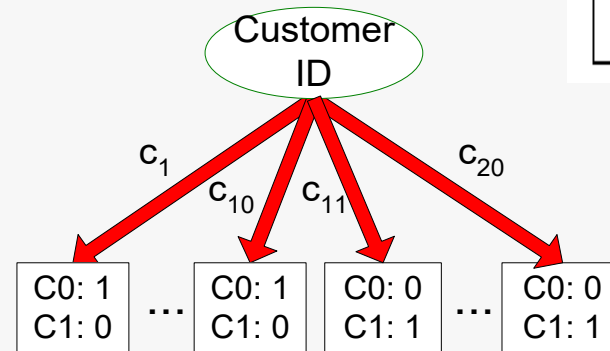
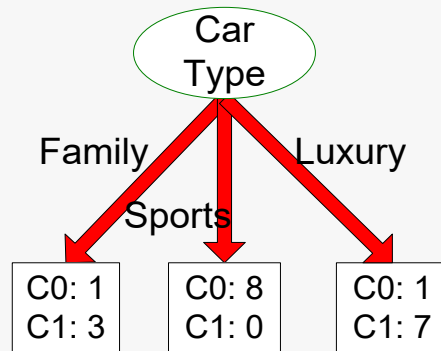
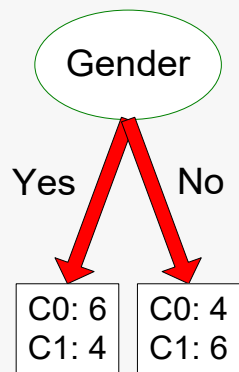
- **Real-world** data tend to be **noisy**.
- Noisy data is data with a large amount of additional meaningless information in it called noise.
- **Data cleaning** (or data cleansing) routines attempt to smooth out noise while identifying **outliers in the data**.
- Three methods to smoothen the data:
 - **Binning**: smooth a sorted data value by consulting its “neighborhood”, that is, the values around it.
 - Regression
 - Outlier analysis

Which test condition is the best?

How to determine the Best Split

Before Splitting: 10 records of class 0,
10 records of class 1

Customer Id	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1



Which test condition is the best?

How to determine the Best Split

I Greedy approach:

- Nodes with **pu^{re}** class distribution are preferred

I Need a measure of node impurity:

C0: 5 C1: 5

High degree of impurity

C0: 9 C1: 1

Low degree of impurity

I Gini Index

$$Gini\ Index = 1 - \sum_{i=0}^{c-1} p_i(t)^2$$

Where $p_i(t)$ is the frequency of class i at node t , and c is the total number of classes

I measures the degree of probability of a particular variable being incorrectly classified when it is chosen randomly.

I Entropy

$$Entropy = - \sum_{i=0}^{c-1} p_i(t) \log_2 p_i(t)$$

I Measures the degree of uncertainty

I Misclassification error

$$Classification\ error = 1 - \max[p_i(t)]$$

- Suppose we have a **binary classification** dataset with **PlayTennis** as the target variable and **Weather** as a feature.

Weather	PlayTennis
Sunny	No
Sunny	No
Overcast	Yes
Rain	Yes
Rain	Yes
Overcast	Yes
Sunny	No
Rain	Yes
Sunny	Yes
Overcast	Yes

Step 1: Calculate Entropy of Parent (Before Split)

$$H(\text{PlayTennis}) = -P(\text{Yes}) \log_2 P(\text{Yes}) - P(\text{No}) \log_2 P(\text{No})$$

- Yes = 6, No = 4 → Total = 10
- $P(\text{Yes}) = \frac{6}{10} = 0.6$, $P(\text{No}) = \frac{4}{10} = 0.4$

$$\begin{aligned} H(\text{PlayTennis}) &= -(0.6 \log_2 0.6 + 0.4 \log_2 0.4) \\ &= -(0.6 \times -0.737 + 0.4 \times -1.322) \\ &= 0.971 \end{aligned}$$

Step 2: Split on Attribute "Weather"

- Sunny (4): [3 No, 1 Yes]**
- Overcast (3): [All Yes]**
- Rain (3): [All Yes]**

- Suppose we have a **binary classification** dataset with **PlayTennis** as the target variable and **Weather** as a feature.

Weather	PlayTennis
Sunny	No
Sunny	No
Overcast	Yes
Rain	Yes
Rain	Yes
Overcast	Yes
Sunny	No
Rain	Yes
Sunny	Yes
Overcast	Yes

Step 3: Compute Weighted Entropy

$$H(Sunny) = - \left(\frac{3}{4} \log_2 \frac{3}{4} + \frac{1}{4} \log_2 \frac{1}{4} \right) = 0.81$$

$$H(Overcast) = 0 \quad (\text{Pure node, all Yes})$$

$$H(Rain) = 0 \quad (\text{Pure node, all Yes})$$

Weighted Entropy:

$$H(Weather) = \left(\frac{4}{10} \times 0.81 \right) + \left(\frac{3}{10} \times 0 \right) + \left(\frac{3}{10} \times 0 \right) = 0.324$$

Step 4: Compute Information Gain

$$IG(Weather) = H(PlayTennis) - H(Weather)$$

$$= 0.971 - 0.324 = 0.647$$

Computing Gini Index of a Single Node

$$\text{Gini Index} = 1 - \sum_{i=0}^{c-1} p_i(t)^2$$

C1	0
C2	6

$$P(C1) = 0/6 = 0 \quad P(C2) = 6/6 = 1$$

$$\text{Gini} = 1 - P(C1)^2 - P(C2)^2 = 1 - 0 - 1 = 0$$

C1	1
C2	5

$$P(C1) = 1/6 \quad P(C2) = 5/6$$

$$\text{Gini} = 1 - (1/6)^2 - (5/6)^2 = 0.278$$

C1	2
C2	4

$$P(C1) = 2/6 \quad P(C2) = 4/6$$

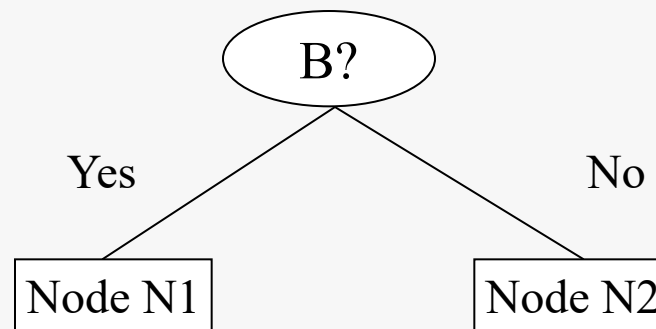
$$\text{Gini} = 1 - (2/6)^2 - (4/6)^2 = 0.444$$

Computing Gini Index of a Single Node

Metric	Entropy (Information Gain)	Gini Index
Measures	Uncertainty (disorder)	Impurity
Formula	$-\sum p_i \log_2 p_i$	$1 - \sum p_i^2$
Value Range	0 (Pure) $\rightarrow$ 1 (Max Disorder)	0 (Pure) $\rightarrow$ 0.5 (Max Disorder)
Decision Trees	Used in ID3, C4.5	Used in CART

Binary Attributes: Computing GINI Index

- Splits into two partitions (child nodes)
- Effect of Weighing partitions:
 - Larger and purer partitions are sought



	Parent
C1	7
C2	5
Gini = 0.486	

$$\begin{aligned}
 \text{Gini}(N1) &= 1 - (5/6)^2 - (1/6)^2 \\
 &= 0.278
 \end{aligned}$$

$$\begin{aligned}
 \text{Gini}(N2) &= 1 - (2/6)^2 - (4/6)^2 \\
 &= 0.444
 \end{aligned}$$

	N1	N2
C1	5	2
C2	1	4
Gini=0.361		

$$\begin{aligned}
 \text{Weighted Gini of N1 N2} &= 6/12 * 0.278 + \\
 &\quad 6/12 * 0.444 \\
 &= 0.361
 \end{aligned}$$

$$\text{Gain} = 0.486 - 0.361 = 0.125$$

Categorical Attributes: Computing Gini Index

Go, change the world

- I For each distinct value, gather counts for each class in the dataset
- I Use the count matrix to make decisions

Multi-way split

	CarType		
	Family	Sports	Luxury
C1	1	8	1
C2	3	0	7
Gini	0.163		

Two-way split
(find best partition of values)

	CarType	
	{Sports, Luxury}	{Family}
C1	9	1
C2	7	3
Gini	0.468	

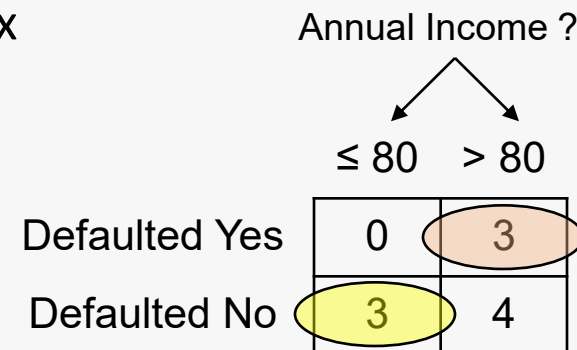
	CarType	
	{Sports}	{Family, Luxury}
C1	8	2
C2	0	10
Gini	0.167	

Which of these is the best?

Continuous Attributes: Computing Gini Index

- | Use Binary Decisions based on one value
- | Several Choices for the splitting value
 - Number of possible splitting values
= Number of distinct values
- | Each splitting value has a count matrix associated with it
 - Class counts in each of the partitions, $A \leq v$ and $A > v$
- | Simple method to choose best v
 - For each v , scan the database to gather count matrix and compute its Gini index
 - Computationally Inefficient! Repetition of work.

ID	Home Owner	Marital Status	Annual Income	Defaulted
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



Continuous Attributes: Computing Gini Index...

Go, change the world

- I For efficient computation: for each attribute,
 - Sort the attribute on values
 - Linearly scan these values, each time updating the count matrix and computing gini index
 - Choose the split position that has the least gini index

Sorted Values →

Cheat	No	No	No	Yes	Yes	Yes	No	No	No	No
	Annual Income									
	60	70	75	85	90	95	100	120	125	220

Continuous Attributes: Computing Gini Index...

Go, change the world

- I For efficient computation: for each attribute,
- Sort the attribute on values
 - Linearly scan these values, each time updating the count matrix and computing gini index
 - Choose the split position that has the least gini index

		Cheat	No	No	No	Yes	Yes	Yes	No	No	No	No				
			Annual Income													
Sorted Values	→		60	70	75	85	90	95	100	120	125	220				
Split Positions	→		55	65	72	80	87	92	97	110	122	172	230			
			<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>

Continuous Attributes: Computing Gini Index...

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Continuous Attributes: Computing Gini Index...

Go, change the world

- I For efficient computation: for each attribute,
- Sort the attribute on values
 - Linearly scan these values, each time updating the count matrix and computing gini index
 - Choose the split position that has the least gini index

		↓																							
	Cheat	No		No		No		Yes		Yes		Yes		No		No		No		No					
		Annual Income																							
Sorted Values	→	60		70		75		85		90		95		100		120		125		220					
Split Positions	→	55		65		72		80		87		92		97		110		122		172		230			
		<=		>		<=		>		<=		>		<=		>		<=		>		<=		>	
	Yes							0	3	1	2														
	No							3	4	3	4														
	Gini							0.343	0.417																

Continuous Attributes: Computing Gini Index...

Go, change the world

- I For efficient computation: for each attribute,
- Sort the attribute on values
 - Linearly scan these values, each time updating the count matrix and computing gini index
 - Choose the split position that has the least gini index

Sorted Values Split Positions		Cheat	No		No		No		Yes		Yes		Yes		No		No		No		No			
			Annual Income																					
	→		60		70		75		85		90		95		100		120		125		220			
	→		55		65		72		80		87		92		97		110		122		172		230	
			<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>
		Yes	0	3	0	3	0	3	0	3	1	2	2	1	3	0	3	0	3	0	3	0	3	0
		No	0	7	1	6	2	5	3	4	3	4	3	4	3	4	4	3	5	2	6	1	7	0
		Gini	0.420		0.400		0.375		0.343		0.417		0.400		<u>0.300</u>		0.343		0.375		0.400		0.420	

- Among these three impurity measurements:
- *Gini* is Used by the CART (classification and regression tree) algorithm for classification trees, and
- *Entropy* is Used by the ID3, C4.5, and C5.0 tree-generation algorithms.

I Entropy at a given node t

$$Entropy = - \sum_{i=0}^{c-1} p_i(t) \log_2 p_i(t)$$

Where $p_i(t)$ is the frequency of class i at node t , and c is the total number of classes

- ◆ Maximum of $\log_2 c$ when records are equally distributed among all classes, implying the least beneficial situation for classification
 - ◆ Minimum of 0 when all records belong to one class, implying most beneficial situation for classification
- Entropy based computations are quite similar to the GINI index computations

Computing Entropy of a Single Node

$$Entropy = - \sum_{i=0}^{c-1} p_i(t) \log_2 p_i(t)$$

C1	0
C2	6

$$P(C1) = 0/6 = 0 \quad P(C2) = 6/6 = 1$$

$$Entropy = - 0 \log 0 - 1 \log 1 = - 0 - 0 = 0$$

C1	1
C2	5

$$P(C1) = 1/6 \quad P(C2) = 5/6$$

$$Entropy = - (1/6) \log_2 (1/6) - (5/6) \log_2 (1/6) = 0.65$$

C1	2
C2	4

$$P(C1) = 2/6 \quad P(C2) = 4/6$$

$$Entropy = - (2/6) \log_2 (2/6) - (4/6) \log_2 (4/6) = 0.92$$

Computing Information Gain After Splitting

I Information Gain:

$$Gain_{split} = Entropy(p) - \sum_{i=1}^k \frac{n_i}{n} Entropy(i)$$

Parent Node, p is split into k partitions (children)

n_i is number of records in child node i

- Choose the split that achieves most reduction (maximizes GAIN)
- Used in the ID3 and C4.5 decision tree algorithms
- Information gain is the mutual information between the class variable and the splitting variable

I Classification error at a node t

- Maximum of $1 - \frac{1}{c}$ when records are equally distributed among all classes, implying the least interesting situation
- Minimum of 0 when all records belong to one class, implying the most interesting situation

Computing Error of a Single Node

$$Error(t) = 1 - \max_i [p_i(t)]$$

C1	0
C2	6

$$P(C1) = 0/6 = 0 \quad P(C2) = 6/6 = 1$$

$$Error = 1 - \max(0, 1) = 1 - 1 = 0$$

C1	1
C2	5

$$P(C1) = 1/6 \quad P(C2) = 5/6$$

$$Error = 1 - \max(1/6, 5/6) = 1 - 5/6 = 1/6$$

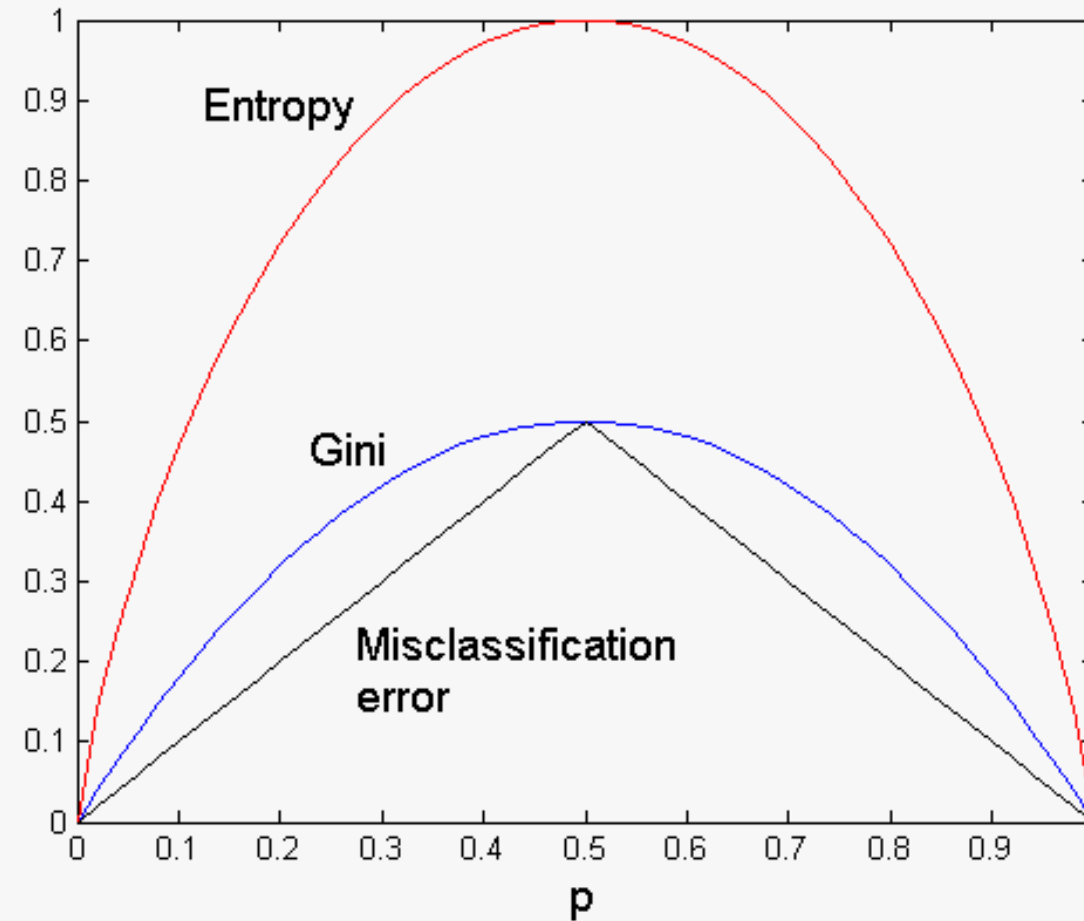
C1	2
C2	4

$$P(C1) = 2/6 \quad P(C2) = 4/6$$

$$Error = 1 - \max(2/6, 4/6) = 1 - 4/6 = 1/3$$

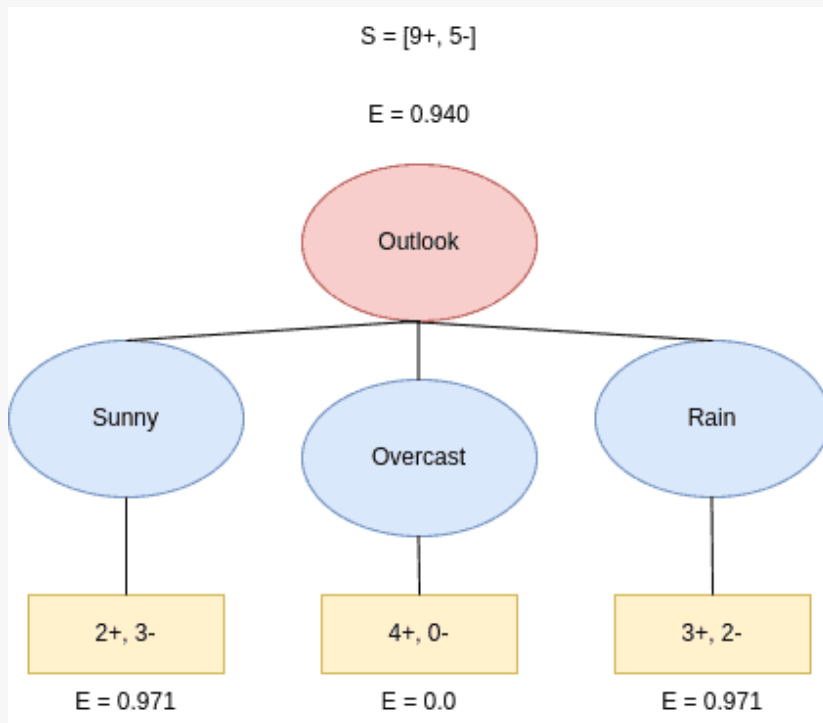
Comparison among Impurity Measures

For a 2-class problem:



Day	Outlook	Temperature	Humidity	Wind	Play Tennis?
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Weak	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Strong	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

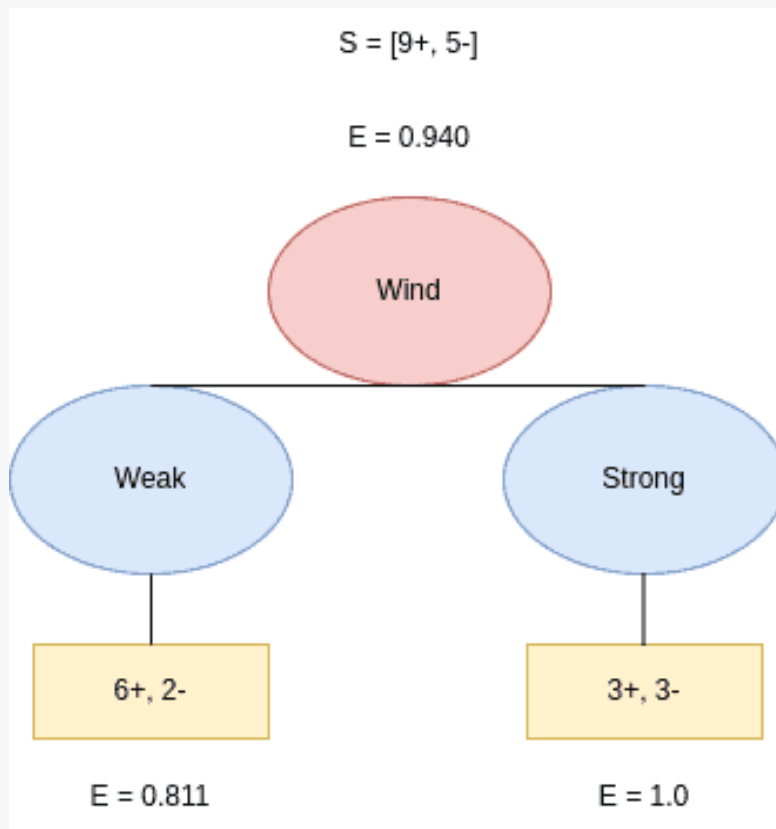
- The dataset has 9 +ve and 5 -ve thus making dataset has $S = [+9 \quad -5]$
- Let's see what happens if we split by *Outcome*. *Outlook* has three different values: *Sunny*, *Overcast*, and *Rain*.
- The possibility of splits are : *sunny* = $[2+, 3-]$, *overcast* = $[4+, 0-]$, *rainy* = $[3+, 2-]$



$$\text{Information Gain} = 0.940 - 5/14(0.970) - 4/14(0) - 5/14(0.971) = 0.247$$

Now split the tree based on wind feature – **weak** and **strong** = $[6+, 2-]$ and $[3+, 3-]$

Information Gain = $0.940 - 8/14(0.811) - 6/14(1) = 0.048$ we calculate the gains for *Humidity* and *Temperature* as well.



$$Gain(S, Humidity) = 0.151$$
$$Gain(S, Temperature) = 0.029$$

We see that the feature with the maximum Information Gain is *Outlook*. **Thus, we conclude that Outlook is the best attribute to split the data at the tree's root.**

We can split the data by the Gini Index too. Let's compute the required probabilities:

$$P(Outlook = Sunny) = \frac{5}{14}, P(Outlook = Overcast) = \frac{4}{14}, P(Outlook = Rain) = \frac{5}{14}$$

- Out of the 14 days in the above example, *Sunny*, *Overcast*, and *Rain* occur 5, 4, and 5 times, respectively.
- Then, we compute the probabilities of a *Sunny* day and playing tennis or not.
- Out of the 5 times when *Outlook=Sunny*, we played tennis on 2 and didn't play it on 3 days:

$$P(Outlook = Sunny \& Play Tennis = Yes) = \frac{2}{5}$$

$$P(Outlook = Sunny \& Play Tennis = No) = \frac{3}{5}$$

Having calculated the required probabilities, we can compute the Gini Index of *Sunny*:

$$gini(Outlook = Sunny) = 1 - \left(\left(\frac{2}{5} \right)^2 + \left(\frac{3}{5} \right)^2 \right) = 0.48$$

We follow the same steps for *Overcast* and *Rain*:

$$P(Outlook = Overcast \ \& \ Play \ Tennis = Yes) = \frac{4}{4}$$

$$P(Outlook = Overcast \ \& \ Play \ Tennis = No) = \frac{0}{4}$$

$$gini(Outlook = Overcast) = 1 - \left(\left(\frac{4}{4} \right)^2 + \left(\frac{0}{4} \right)^2 \right) = 0$$

$$P(Outlook = Rain \ \& \ Play \ Tennis = Yes) = \frac{3}{5}$$

$$P(Outlook = Rain \ \& \ Play \ Tennis = No) = \frac{2}{5}$$

$$gini(Outlook = Rain) = 1 - \left(\left(\frac{3}{5} \right)^2 + \left(\frac{2}{4} \right)^2 \right) = 0.48$$

Therefore, the weighted sum $gini(Outlook)$ of the above Gini Indices is:

$$gini(Outlook) = \frac{5}{14} \cdot 0.48 + \frac{4}{14} \cdot 0 + \frac{5}{14} \cdot 0.48 = 0.343$$

Similarly, we compute the Gini Index of *Temperature*, *Humidity*, and *Wind*:

$$gini(Temperature) = 0.423, \quad gini(Humidity) = 0.368, \quad gini(Wind) = 0.5$$

We conclude that *Outlook* has the lowest Gini Index. Hence, we'll choose it as the root node.

Example: DT

Go, change the world

Problem Statement: A company wants to classify whether a customer will **purchase a product** based on the following features: **Age** and **Income Level**.

Customer	Age	Income	Purchase (Yes/No)
1	Young	Low	No
2	Young	Medium	No
3	Young	High	Yes
4	Middle	Low	Yes
5	Middle	Medium	Yes
6	Middle	High	No
7	Senior	Low	Yes
8	Senior	Medium	Yes
9	Senior	High	No

We need to build a **Decision Tree** to classify new customers based on **Age** and **Income**.

Example: DT

Go, change the world

Step 1: Compute Entropy for the Target Variable (Purchase)

Entropy formula:

$$H(S) = - \sum p_i \log_2(p_i)$$

Counting "Yes" and "No" for Purchase

- Yes = 5
- No = 4
- Total = 9

$$H(S) = - \left(\frac{5}{9} \log_2 \frac{5}{9} + \frac{4}{9} \log_2 \frac{4}{9} \right)$$

Using $\log_2 5/9 \approx -0.847$ and $\log_2 4/9 \approx -1.169$:

$$H(S) = - \left(\frac{5}{9} \times (-0.847) + \frac{4}{9} \times (-1.169) \right)$$

$$H(S) = 0.991$$



Examp

Step 3: Compute Entropy and Information Gain for "Income"

Split the dataset by Income

Low (3 instances)

- Yes = 2, No = 1

$$H(Low) = 0.918$$

Medium (3 instances)

- Yes = 2, No = 1

$$H(Medium) = 0.918$$

High (3 instances)

- Yes = 1, No = 2

$$H(High) = 0.918$$

Weighted Entropy for Income

$$H(Income) = \frac{3}{9}(0.918) + \frac{3}{9}(0.918) + \frac{3}{9}(0.918) = 0.918$$

Information Gain for Income

$$IG(Income) = 0.991 - 0.918 = 0.073$$

Step 2: Compute Entropy and Information Gain for "Age"

Split the dataset by Age

Young (3 instances)

- Yes = 1, No = 2

$$H(Young) = - \left(\frac{1}{3} \log_2 \frac{1}{3} + \frac{2}{3} \log_2 \frac{2}{3} \right)$$

$$H(Young) = - (0.333 \times (-1.585) + 0.667 \times (-0.585))$$

$$H(Young) = 0.918$$

Middle (3 instances)

- Yes = 2, No = 1

$$H(Middle) = - \left(\frac{2}{3} \log_2 \frac{2}{3} + \frac{1}{3} \log_2 \frac{1}{3} \right) = 0.918$$

Senior (3 instances)

- Yes = 2, No = 1

$$H(Senior) = 0.918$$

Weighted Entropy for Age

$$H(Age) = \frac{3}{9}H(Young) + \frac{3}{9}H(Middle) + \frac{3}{9}H(Senior)$$

$$H(Age) = \frac{3}{9}(0.918) + \frac{3}{9}(0.918) + \frac{3}{9}(0.918) = 0.918$$

Information Gain for Age

$$IG(Age) = H(S) - H(Age)$$

$$IG(Age) = 0.991 - 0.918 = 0.073$$

Example: DT

Go, change the world

Step 4: Compute Gini Index for "Age" and "Income"

Gini Index formula:

$$Gini(S) = 1 - \sum p_i^2$$

$$Gini(S) = 1 - \left(\frac{5^2}{9} + \frac{4^2}{9} \right)$$

$$Gini(S) = 0.493$$

$$Gini(Age) = 0.444, \quad Gini(Income) = 0.444$$

Both Age and Income have the same Information Gain and Gini Index. We can choose either as the first split.

```

      [ Age ]
      /   |   \
    Young Middle Senior
    / | \   / | \   / | \
  Low Med High Low Med High Low Med High
No  No  Yes Yes Yes No  Yes Yes No

```



Decision Tree Numerical Problem- Dataset: Loan Approval Prediction

Go, change the world

ID	Income	CreditScore	Student	AgeGroup	LoanApproved
1	High	Good	No	Young	Yes
2	High	Good	No	Middle	Yes
3	Medium	Good	No	Young	Yes
4	Low	Fair	Yes	Young	No
5	Low	Fair	Yes	Senior	No
6	Low	Good	Yes	Senior	Yes
7	Medium	Fair	Yes	Middle	Yes
8	High	Fair	No	Young	No
9	High	Fair	Yes	Senior	Yes
10	Low	Good	Yes	Middle	Yes
11	Medium	Good	Yes	Senior	Yes
12	Medium	Fair	No	Middle	No
13	High	Good	Yes	Young	Yes
14	Low	Fair	No	Senior	No



Decision Tree Numerical Problem- Dataset: Loan Approval Prediction

- **Part A: Entropy** - Compute entropy of **LoanApproved**.
- **Part B: Information Gain:** Income , CreditScore , Student , AgeGroup
- **Part C: Root Selection:** Select the best root node.
- **Part D: Build Decision Tree** - Construct the decision tree (at least 2 levels).
- **Part E: Prediction (IMPORTANT)**
- Predict the class for the following new cases:

Income	CreditScore	Student	AgeGroup
Low	Fair	Yes	Young
Income	CreditScore	Student	AgeGroup
High	Fair	No	Young
Income	CreditScore	Student	AgeGroup
Medium	Good	Yes	Senior



2. Attribute: Income : Values: High, Medium, Low

- **High (4 records):** Yes = 3, No = 1, $H = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4}$
- $= -(0.75)(-0.41) - (0.25)(-2) = 0.31 + 0.5 = 0.81$
- **Medium (4 records):** Yes = 3, No = 1 $\rightarrow$ Same as above: $H = 0.81$
- **Low (6 records):** Yes = 3, No = 3, $H = -0.5 \log_2 (0.5) - 0.5 \log_2 (0.5) = 1$
- **Weighted Entropy**
- $= \frac{4}{14} (0.81) + \frac{4}{14} (0.81) + \frac{6}{14} (1)$
- $= 0.231 + 0.231 + 0.429 = 0.891$
- **Information Gain:** $IG = 0.94 - 0.89 = 0.05$: **IG(Income) $\approx$ 0.05**

3. Attribute: Student: Values: Yes, No

- Student = Yes (8 records): Yes = 6, No = 2

- $H = -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8}$

- $= -(0.75)(-0.41) - (0.25)(-2) = 0.81$

Student = No (6 records): Yes = 3, No = 3, $H = 1$

Weighted Entropy: $= \frac{8}{14} (0.81) + \frac{6}{14} (1) = 0.463 + 0.429 = 0.892$

- Information Gain $IG = 0.94 - 0.892 = 0.048$ **IG(Student) ≈ 0.05**



■ Branch 2: CreditScore = Fair

Go, change the world

■ Now consider subset:

Income	Student	Output
Low	Yes	No
Low	Yes	No
High	No	No
Medium	No	No
High	Yes	Yes
Medium	Yes	Yes



Case 1

- Low, Fair, Yes, Young
- Path: CreditScore = Fair , Income = Low
- **Prediction = NO**

Case 2

- High, Fair, No, Young
- Path: CreditScore = Fair , Income = High , Student = No
- **Prediction = NO**

Case 3

- Medium, Good, Yes, Senior
 - Path: CreditScore = Good
 - **Prediction = YES**
-



- Non parametric approach - doesn't require any **prior assumptions** like type of **probability distribution**
- Optimal - DT is a **NP problem** - no optimal solution
- Techniques developed for constructing decision trees are computationally **inexpensive**, making it possible to **quickly construct models** even when the training set size is very large.
- Decision trees, especially **smaller-sized trees**, are relatively **easy to interpret**.
- Decision tree algorithms are quite **robust** to the presence of **noise**, especially when methods for avoiding overfitting were employed
- The presence of **redundant attributes** does **not adversely affect** the **accuracy** of decision trees - An attribute is redundant if it is strongly correlated with another attribute in the data.
- Studies have shown that the choice of **impurity measure** has little **effect** on the **performance of decision tree** induction algorithms.

Decision Tree Based Classification

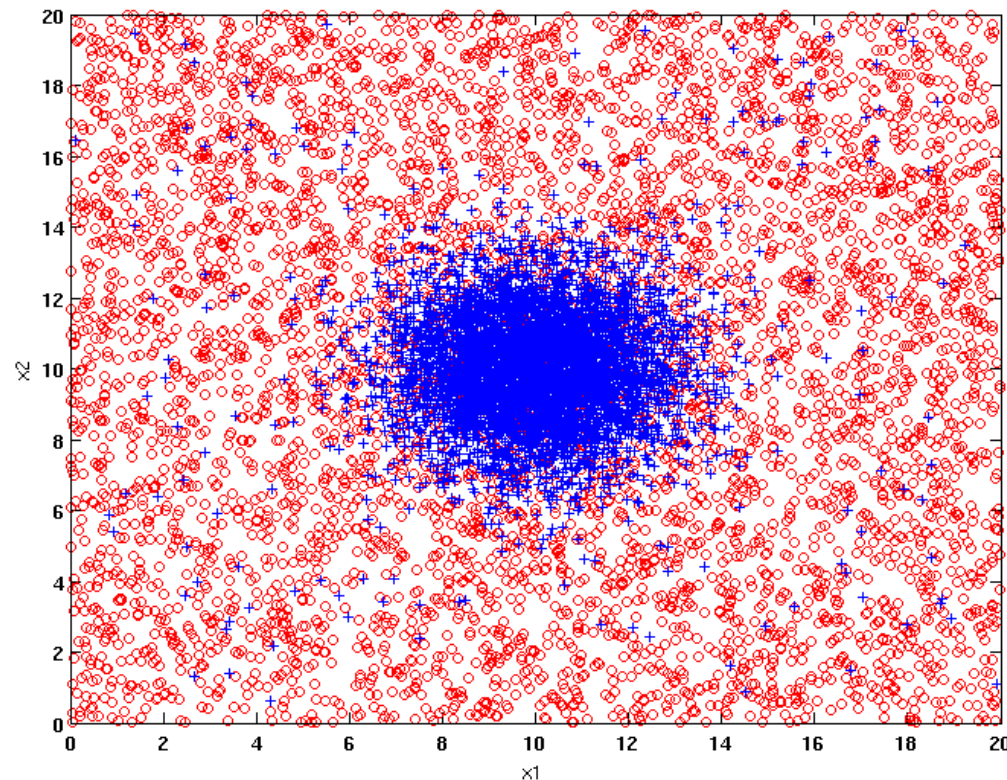
Go, change the world

I Advantages:

- Relatively inexpensive to construct
- Extremely fast at classifying unknown records
- Easy to interpret for small-sized trees
- Robust to noise (especially when methods to avoid overfitting are employed)
- Can easily handle redundant attributes
- Can easily handle irrelevant attributes (unless the attributes are **interacting**)

I Disadvantages: .

- Due to the greedy nature of splitting criterion, **interacting** attributes (that can distinguish between classes together but not individually) may be passed over in favor of other attributed that are less discriminating.
- Each decision boundary involves only a single attribute



Two class problem:

+ : 5400 instances

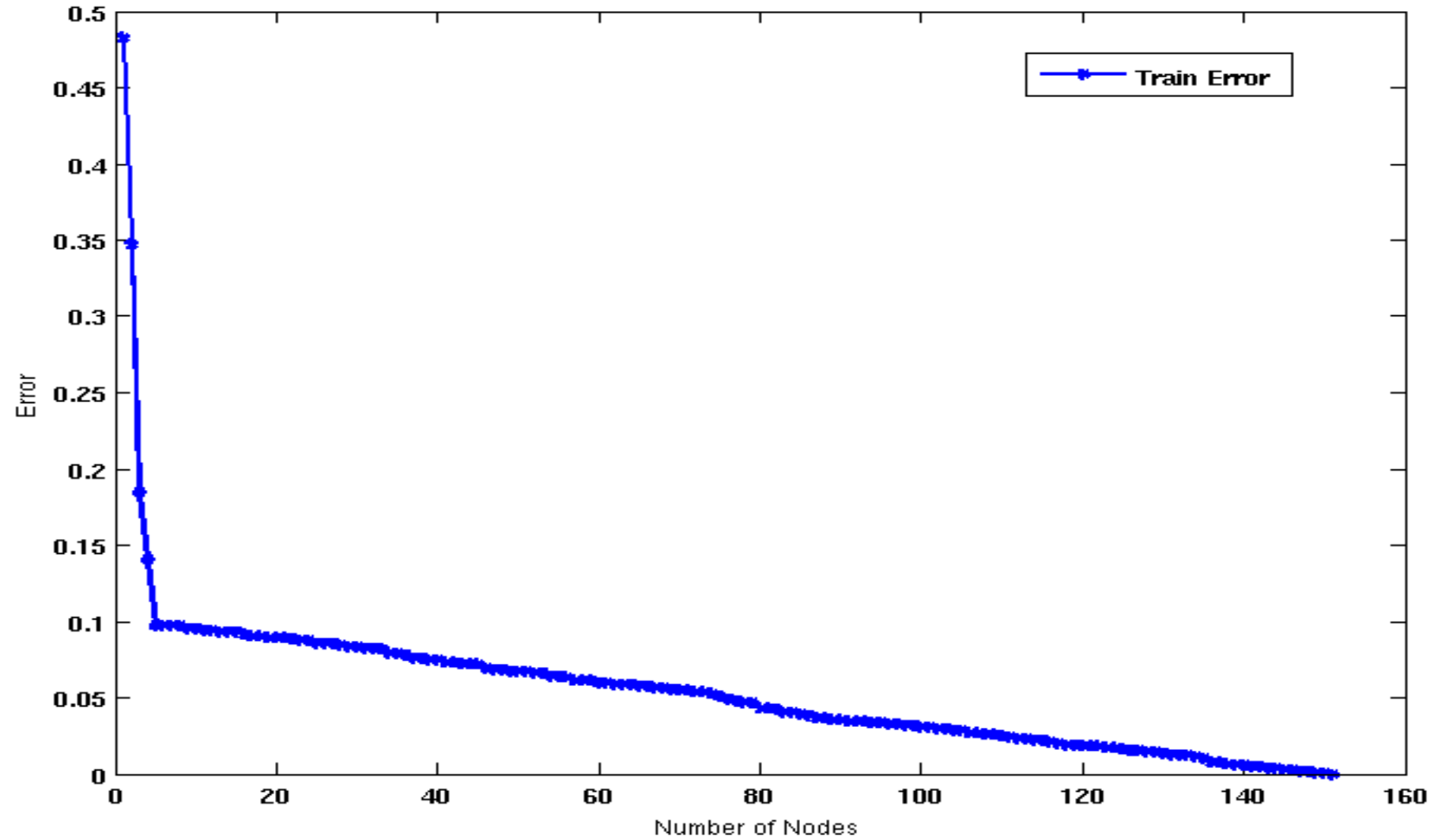
- 5000 instances generated from a Gaussian centered at (10,10)
- 400 noisy instances added

o : 5400 instances

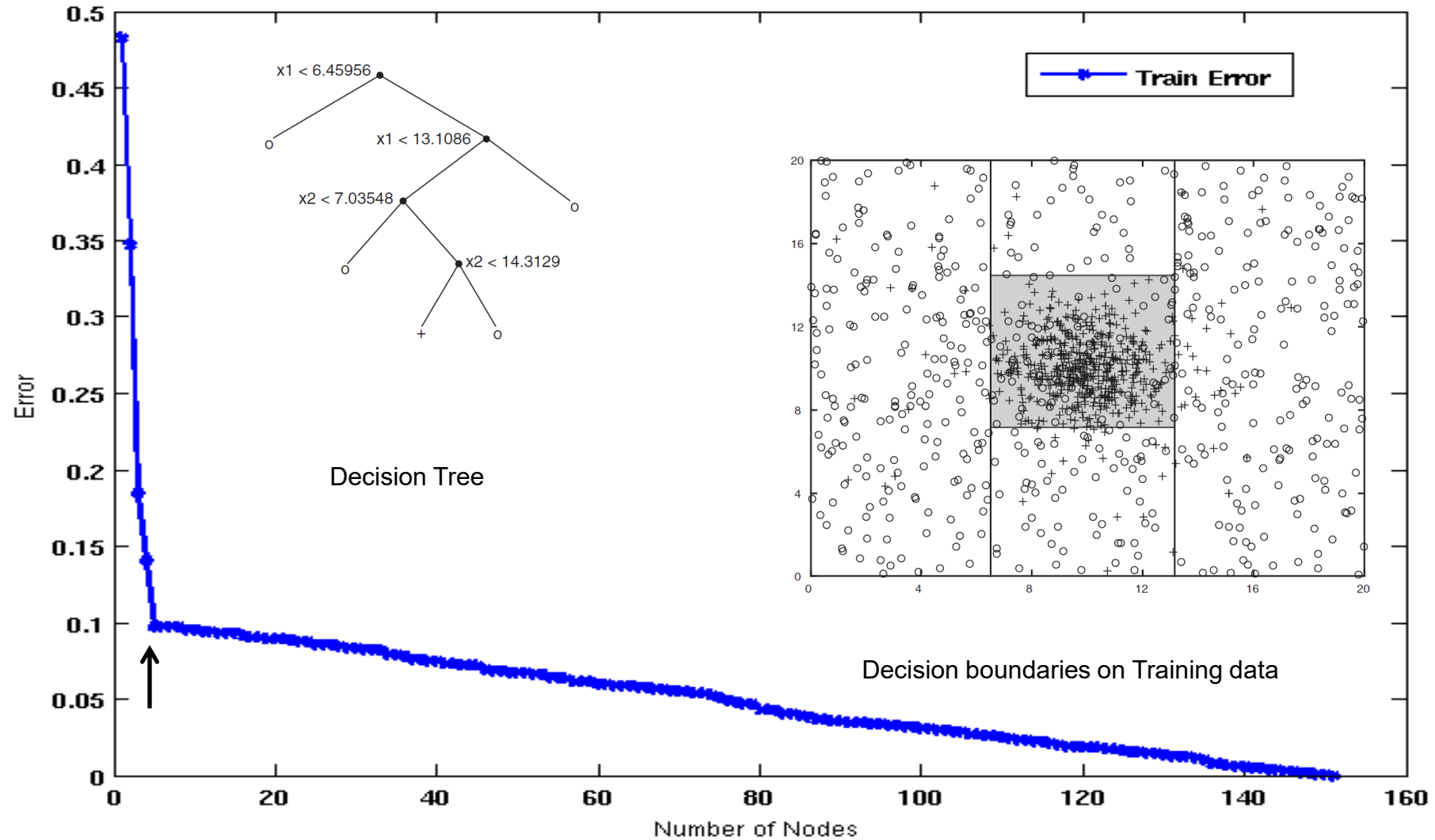
- Generated from a uniform distribution

10 % of the data used for training and 90% of the data used for testing

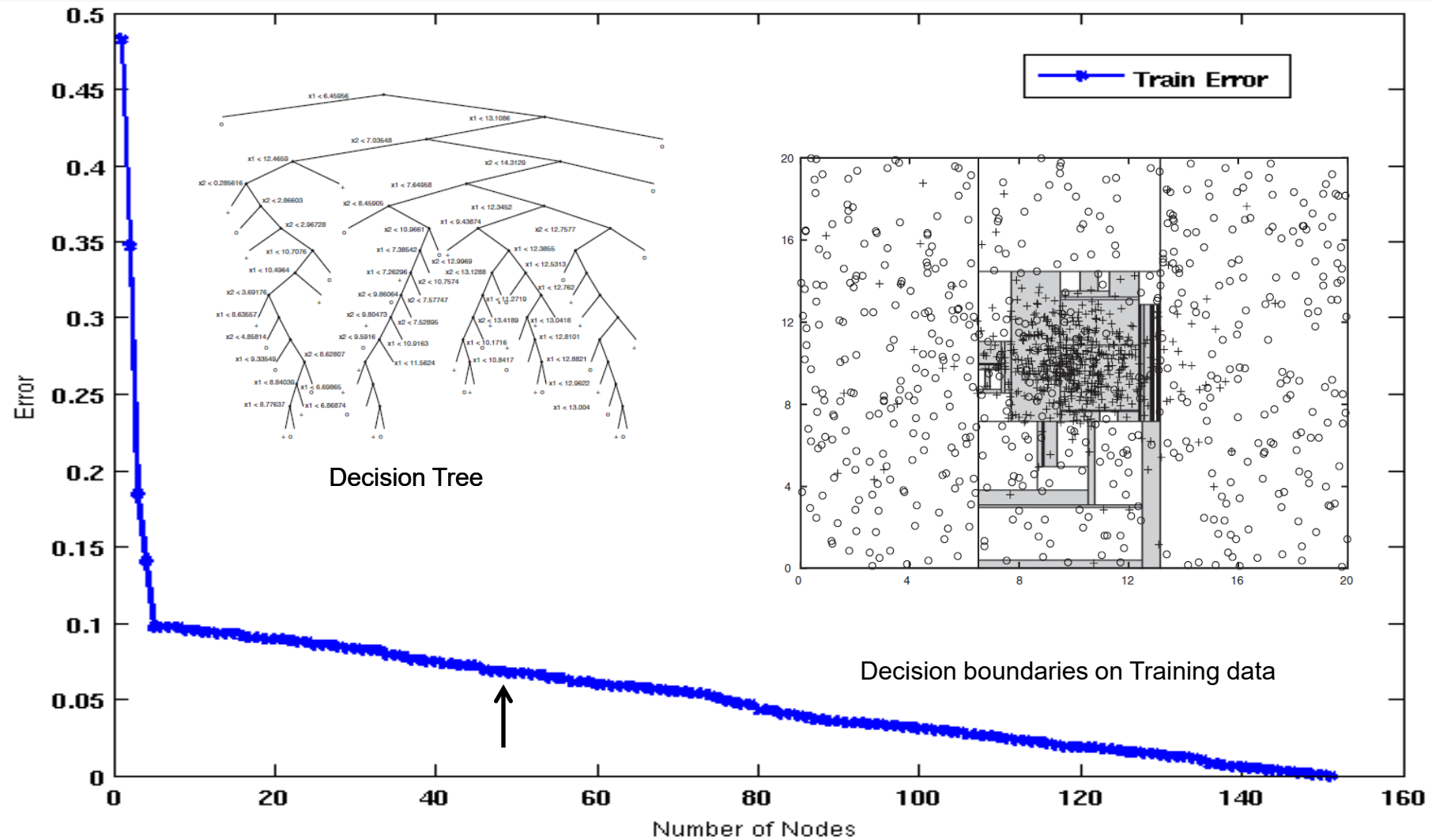
Increasing number of nodes in Decision Trees



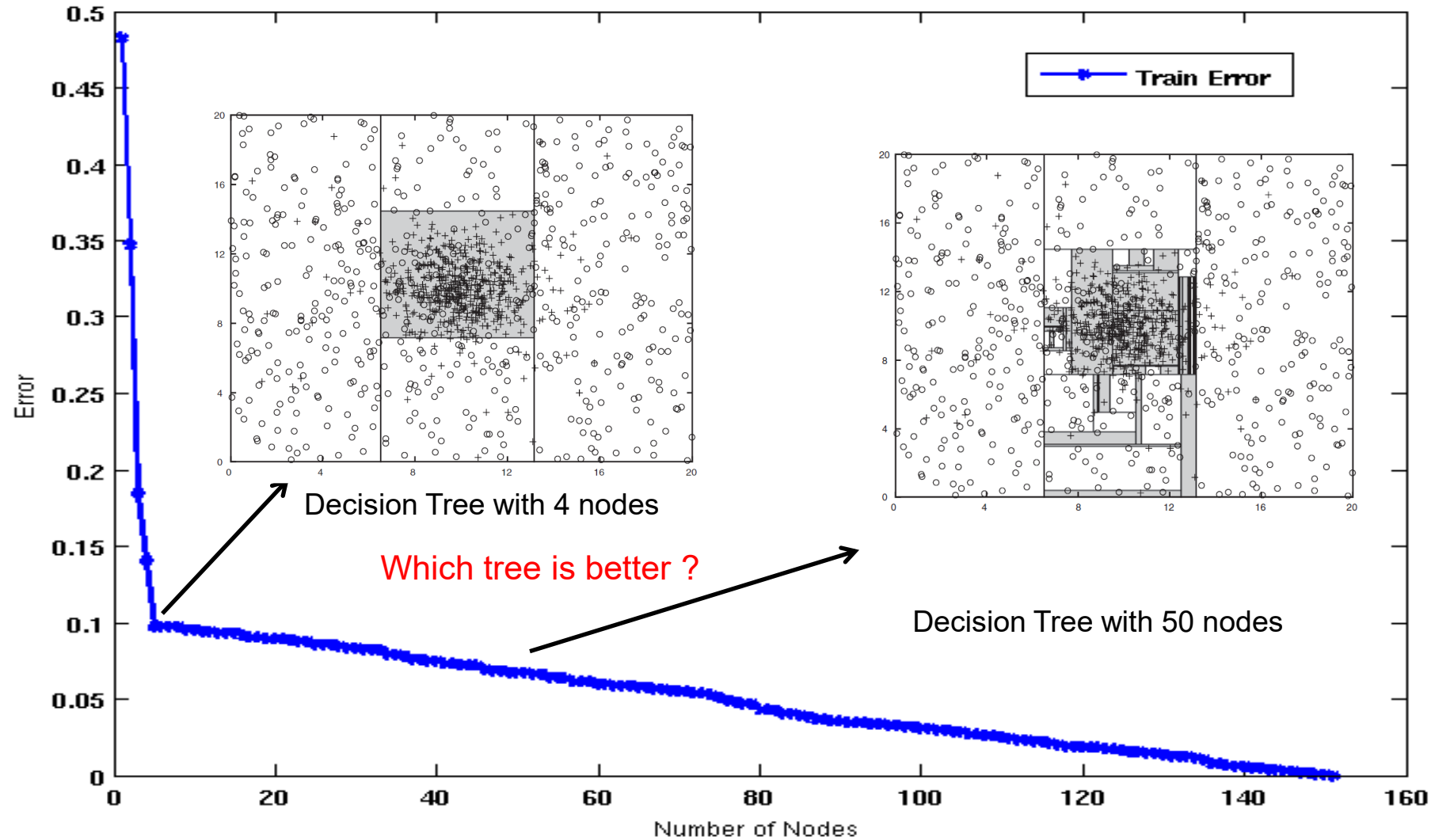
Decision Tree with 4 nodes

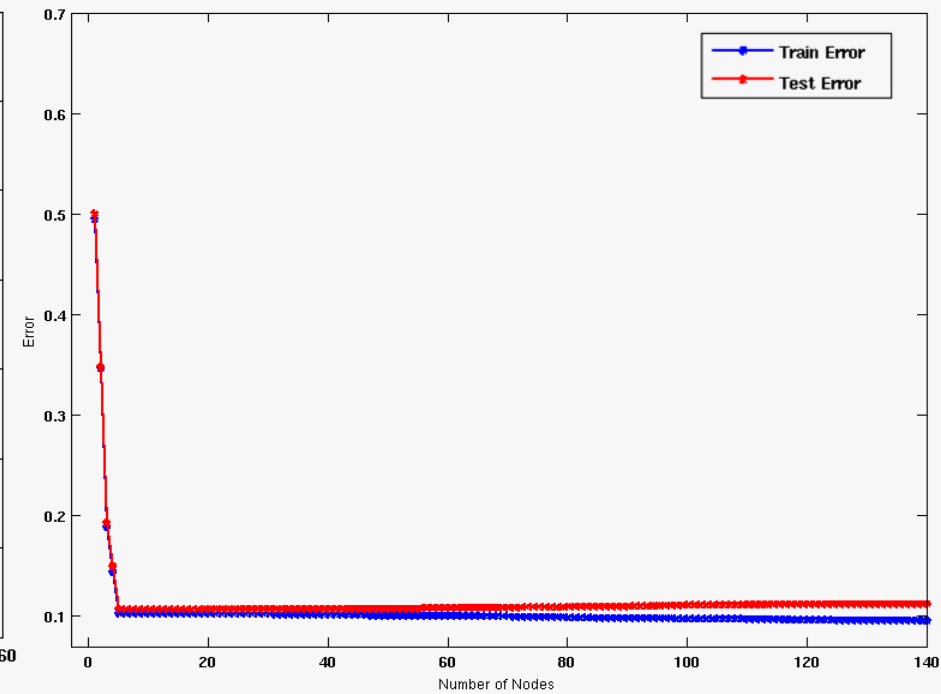
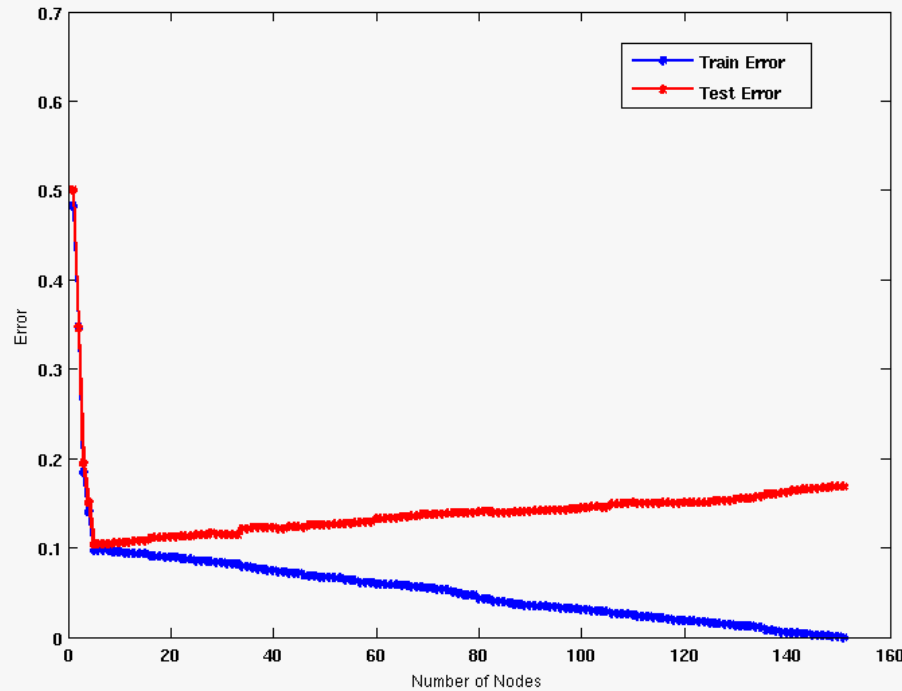


Decision Tree with 50 nodes



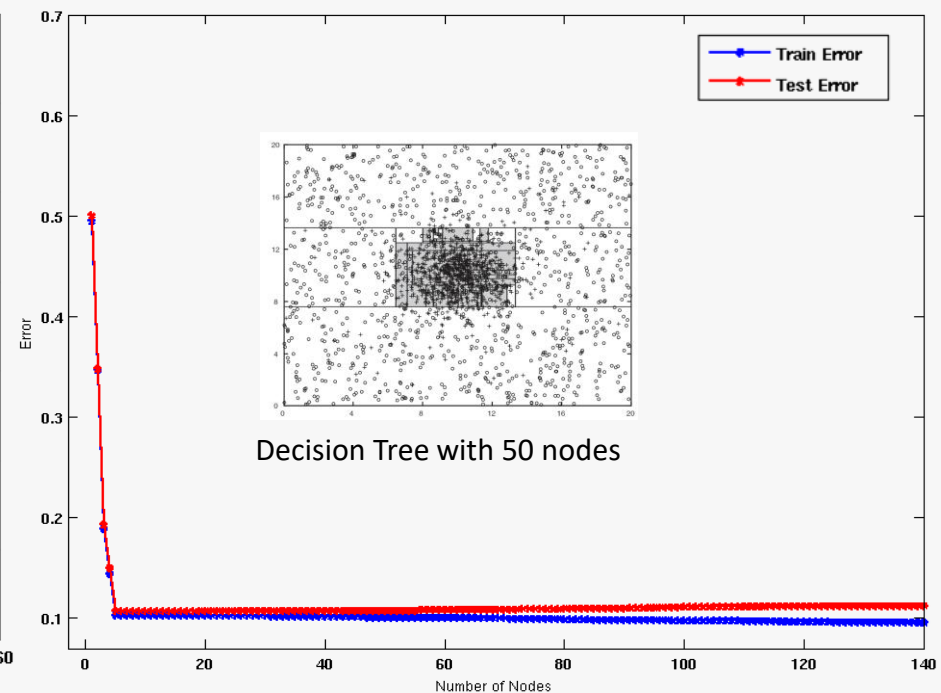
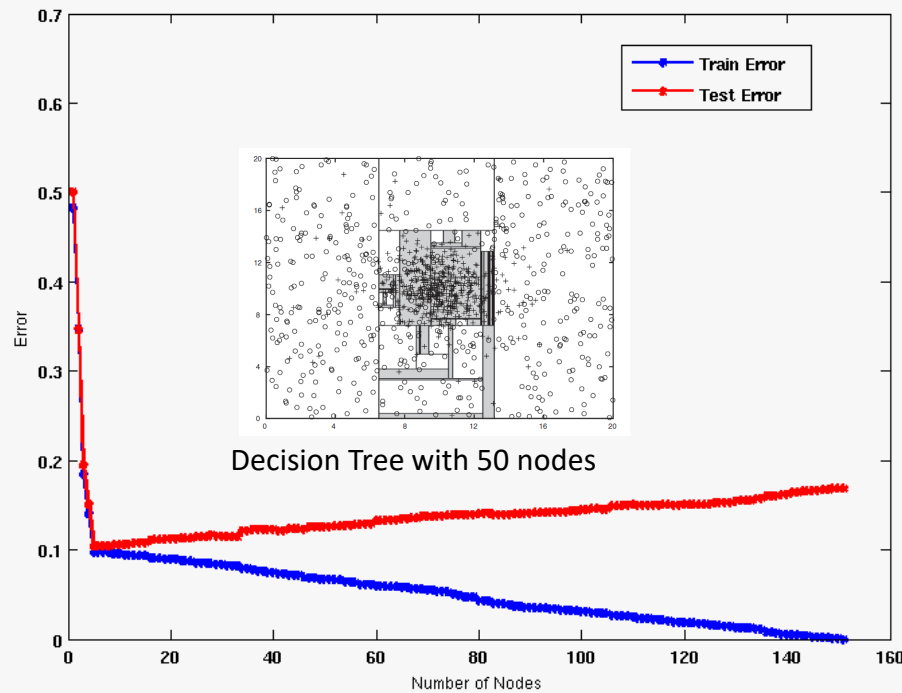
Which tree is better?





Using twice the number of data instances

- Increasing the size of training data reduces the difference between training and testing errors at a given size of model



Using twice the number of data instances

- Increasing the size of training data reduces the difference between training and testing errors at a given size of model



-
- Not enough training data
 - High model complexity
 - Multiple Comparison Procedure
-

■ What is Overfitting?

- Overfitting occurs when a **model fits too closely to the training data** and may become **less accurate** when **encountering new data** or predicting future outcomes.
 - In an overfit condition, a model memorizes the **noise of the training data** and fails to capture essential patterns
 - In decision trees, In order to fit the data (even noisy data), the model keeps generating new nodes and ultimately the tree becomes too complex to interpret.
-



Underfitting: When a model has not learned the patterns in the training data well and is unable to generalize well on the new data, it is known as underfitting.

An underfit model has poor performance on the training data and will result in unreliable predictions.

Reasons for Underfitting:

1. High bias and low variance
2. The size of the training dataset used is not enough.
3. The model is too simple.
4. Training data is not cleaned and also contains noise in it.

Techniques to reduce underfitting:

1. Increase model complexity
 2. Increase the number of features, performing feature engineering
 3. Remove noise from the data.
 4. Increase the number of epochs or increase the duration of training to get better results.
-

- **Overfitting:** A statistical model is said to be overfitted when the model does not make accurate predictions on testing data.
- When a model gets trained with so much data, it starts learning from the noise and inaccurate data entries in our data set. And when testing with test data results in High variance.
- Then the model does not categorize the data correctly, because of too many details and noise.

Reasons for Overfitting are as follows:

1. High variance and low bias
2. The model is too complex
3. The size of the training data

Techniques to reduce overfitting:

1. Increase training data.
2. Reduce model complexity.

Bias

- ⑩ Bias is difference between the predicted value and actual value.
 - ⑩ Bias is one type of error which occurs due to wrong assumptions about data such as assuming data is linear when in reality, data follows a complex function.
 - ⑩ Occurs when model pays little attention to the underlying data. Even if we provide more training samples a low bias model does not change.
 - ⑩ **It is basically underfitting of data.**
 - ⑩ Linear regression, logistic regression etc suffer from low bias
-

Variance

- ⑩ Variance is the difference in the model prediction on the train and test data.
 - ⑩ It is the difference between many model's predictions (fluctuating)
 - ⑩ If the model performs well on the training data (low training error) but fails to generalize well to the test data (high test dataset error), it is said to have a high variance.
 - ⑩ This essentially means that your model is overfitting the training data.
-
- ⑩ It can be because the model captures random noise in the data or it has not

- ⑩ To decrease the bias we need to increase the model complexity, which means the variance increases.
- ⑩ To lower the variance means the model may not fit the data well and hence the bias increases.
- ⑩ An optimal model is one that n

two.

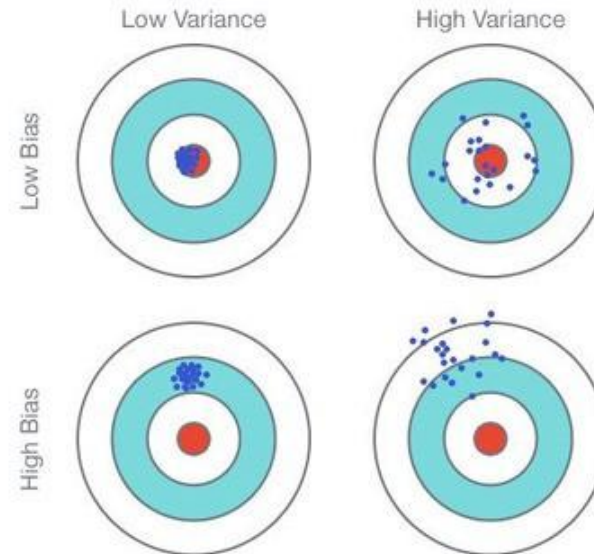
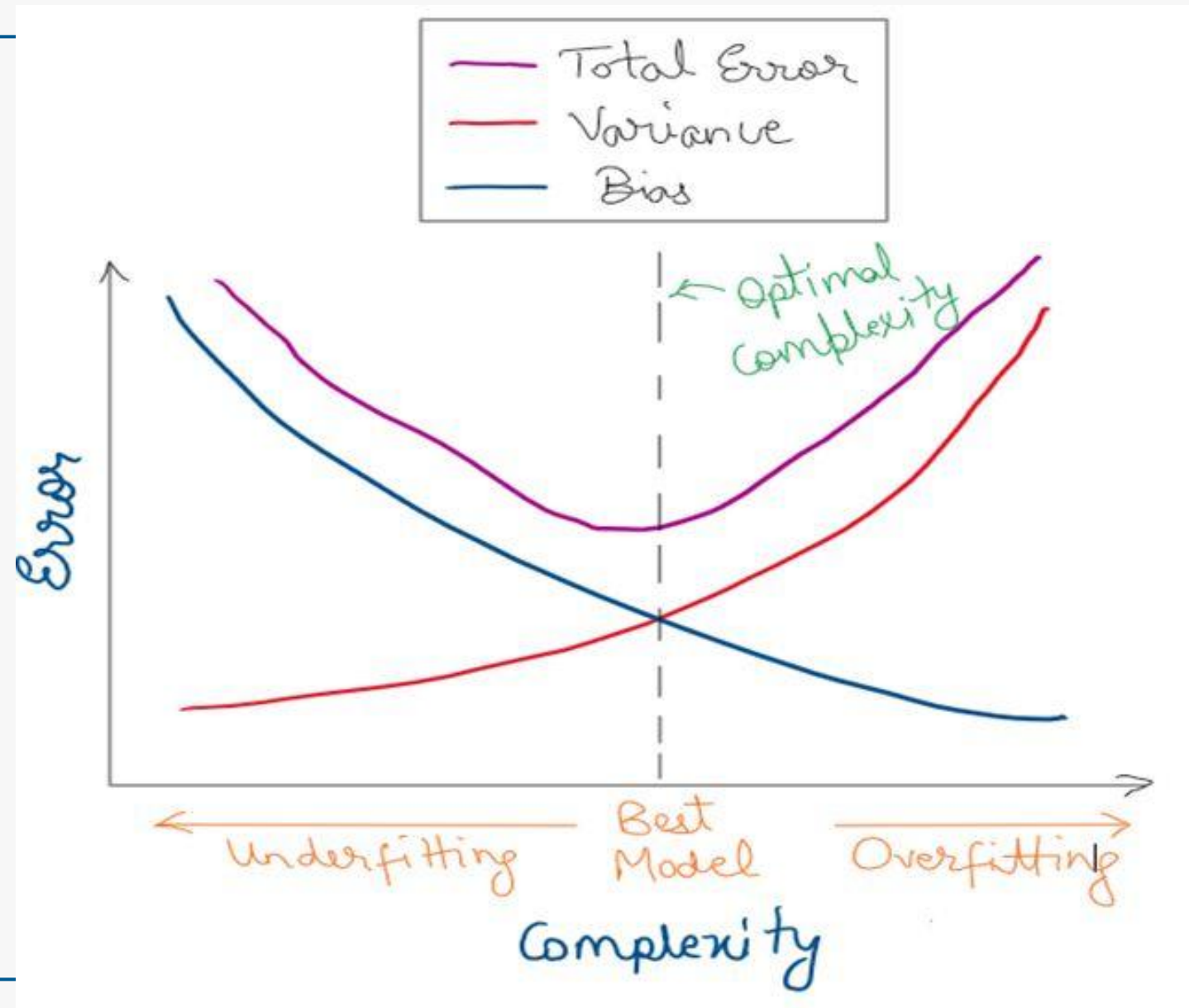


Fig. 1: Graphical Illustration of bias-variance trade-off, Source: Scott Fortmann-Roe., Understanding Bias-Variance Trade-off

Bias - Variance Dilemma

Go, change the world



- Techniques to overcome overfitting:

- Post-Pruning:

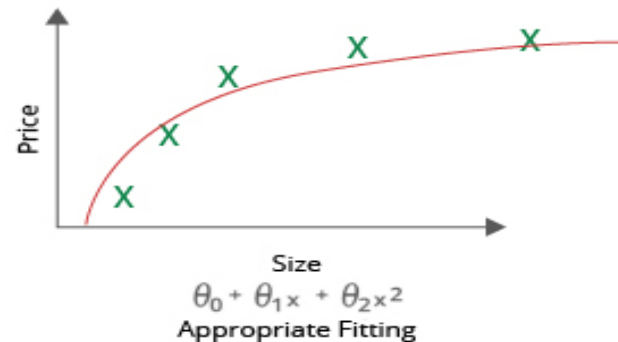
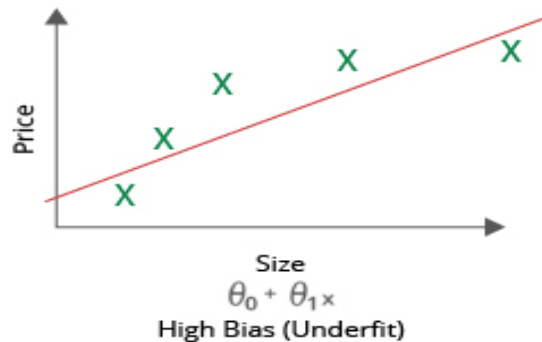
- Post-Pruning or ‘backward pruning’ is a technique that **eliminates branches from a “completely grown” decision tree** model to reduce its complexity and variance.
 - This technique allows the **decision tree to grow to its full depth**, then **removes branches to prevent the model from overfitting**.
 - By doing so, the model might slightly **increase the training error** but drastically **decrease the testing error**

■ Techniques to overcome overfitting:

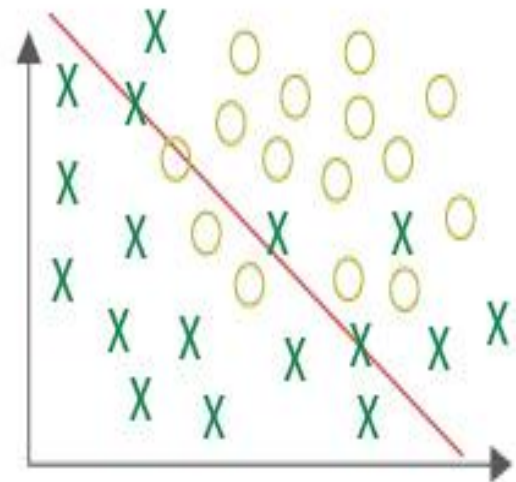
- **Pruning** is a technique that **removes parts of the decision tree** and prevents it from growing to its full depth.
- **Pre-Pruning**, also known as ‘**Early Stopping**’ or ‘**Forward Pruning**’, **stops the growth of the decision tree – preventing it from reaching its full depth.**
- The hyperparameters that can be tuned for pre-pruning or early stopping are
 - **max_depth** (if not specified - nodes are expanded until all leaves are pure or until all leaves contain less than min_samples_split samples.
 - **min_samples_leaf** (samples required at leaf node)
 - **and min_samples_split** - Specifies the minimum number of samples required to split an internal node.

Out of the box discussion - may be useful (Bias)

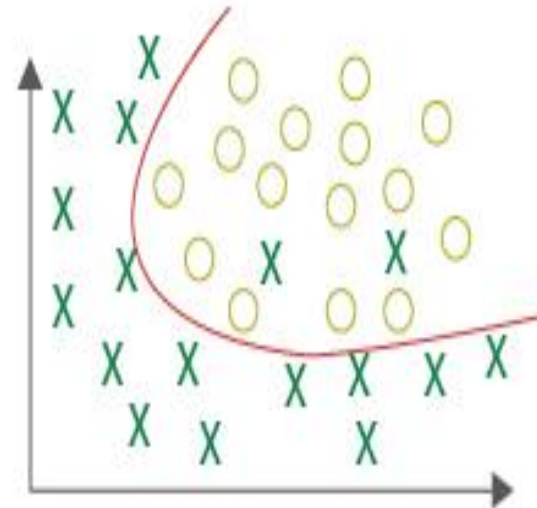
- Bias is difference between the model predicted value and actual value.
- Bias is one type of error which occurs due to wrong assumptions about data such as assuming data is linear when in reality, data follows a complex function.
- Occurs when model pays little attention to the underlying data. Even if we provide more training samples a low bias model does not change.
- **It is basically underfitting of data.**
- Linear regression, logistic regression etc suffer from low bias



Out of the box discussion - may be useful (Bias)



Under-fitting
(too simple to explain the variance)

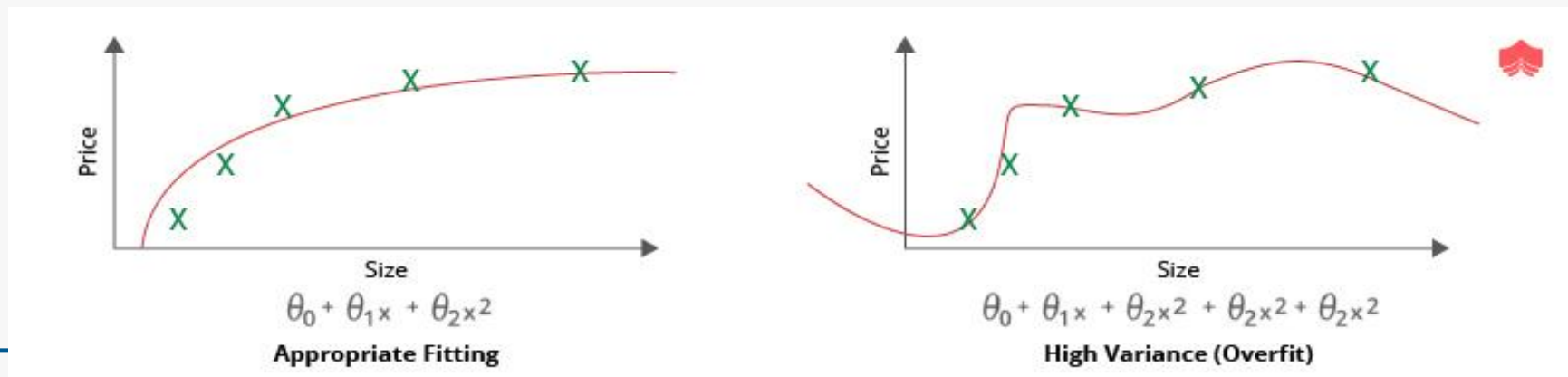


Appropriate-Fitting

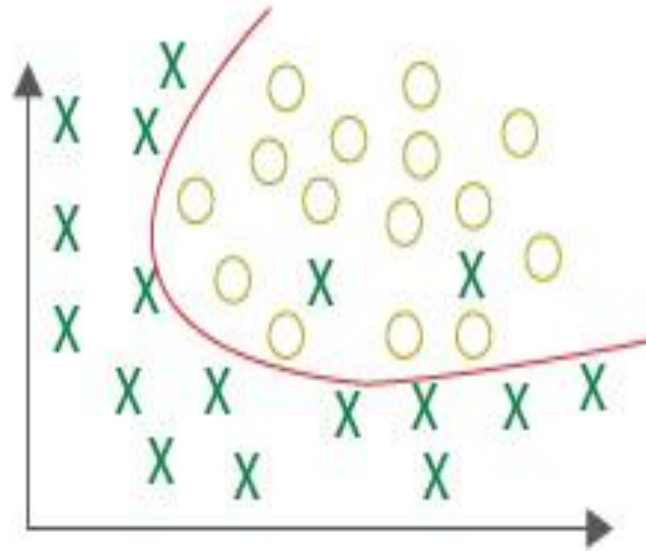


Out of the box discussion - may be useful (Variance)

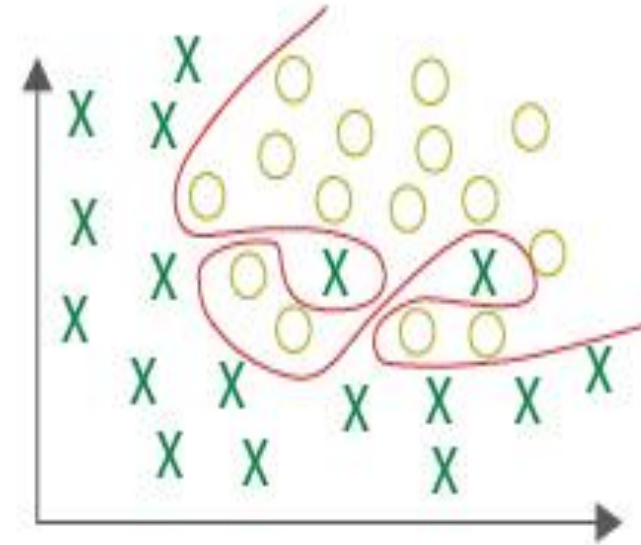
- Variance is the difference in the model prediction on the train and test data.
- **It** is the difference between many model's predictions(fluctuating)
- If the model performs well on the training data (low training error) but fails to generalize well to the test data (high test dataset error), it is said to have a high variance.
- This essentially means that your model is overfitting the training data.
- It can be because the model captures random noise in the data or it has not seen enough data in the training to generalize well to the unseen examples.



Out of the box discussion - may be useful (Variance)



Appropriate-Fitting



Over-Fitting
(Forcefitting --too good to be true)

Out of the box discussion - may be useful Bias - Variance Dilemma

- To **decrease the bias**, we need to **increase the model complexity**, which means the **variance increases**.
- To **lower the variance** means the **model may not fit** the data well and hence the **bias increases**.
- An **optimal model** is one that make good trade-off between two.

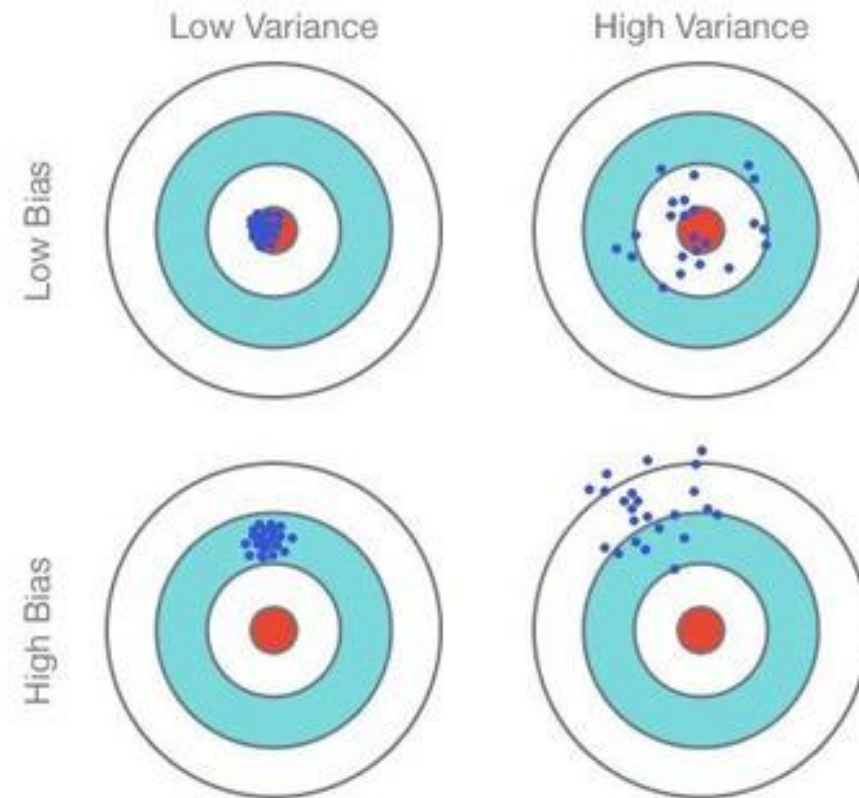
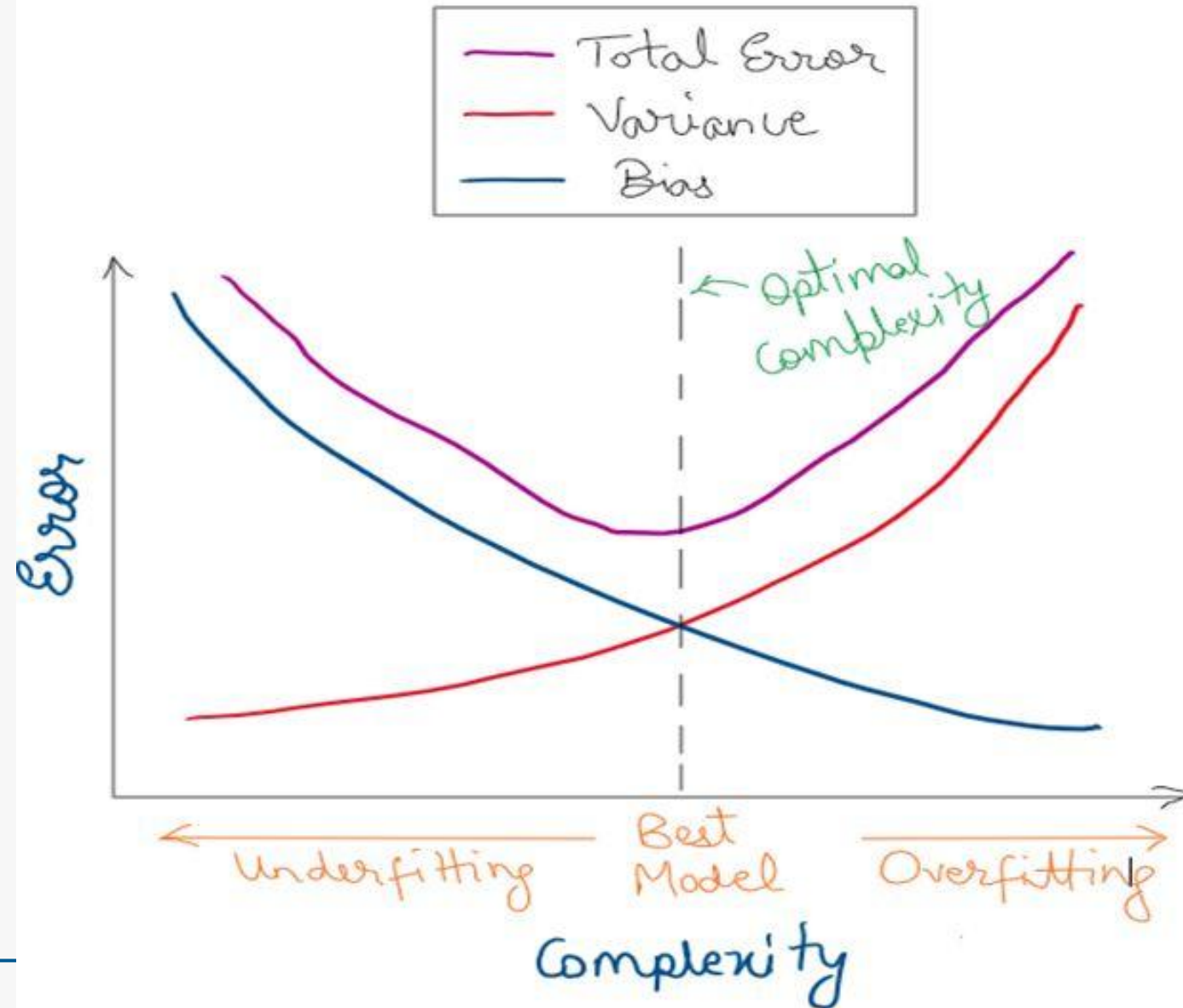


Fig. 1: Graphical Illustration of bias-variance trade-off , Source: Scott Fortmann-Roe., Understanding Bias-Variance Trade-off

Out of the box discussion - may be useful

Change the world

- To avoid **underfitting (high bias)**
 - Try to increase complexity of model
 - Train for longer time
 - Try to increase the number of features by finding new features or making new features from the existing ones.
 - Decrease regularization
 - Use different model
- To avoid **overfitting (high variance)**, try the following –
 - Increase the training data (collecting more data/augment the training dataset)
 - Less complex model
 - Early stop(reduce epocs)
 - Remove features (not recommended)
 - Ensembling
 - L1/L2 regularization to simplify your model.